



Anti-barnacle biofouling coatings for the protection of marine vessels: synthesis and progress

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Abstract

Marine biofouling has gnawed both mobile and non-mobile marine structures since time immemorial, leading to the deterioration of designed operational capabilities as well as a loss of valuable economic revenues. Mitigation of biofouling has been the primary focus of researchers and scientists from across the globe to save billions of dollars wasted due to the biological fouling of marine structures. The availability of an appropriate environment along with favorable substrata initiates biofilm formation within a few minutes. The crucial element in establishing a gelatinous biofilm is the excreted metabolites of destructive nature and exopolymeric substances (EPSs). These help in securing as well as signaling numerous foulants to establish themselves on this substrate. The larvae of various benthic invertebrates adhere to these suitable surfaces and transform from juveniles to adult barnacles depending upon the environment. Despite biofouling being characteristically witnessed for a month or lengthier timeframe, the preliminary phases of the fouling process typically transpire on a much lesser timescale. A few natural and synthetic additives had demonstrated excellent non-toxic anti barnacle establishment capability; however, further development into commercial products is still far-fetched. This review collates the specific anti-barnacle coatings, emphasizing natural additives, their sources of extraction, general life cycle analysis, and concluding future perspectives of this niche product.

Keywords Biofouling · Barnacles · Natural additives · Life cycle assessment (LCA) · Carbon equivalent

Introduction

One of the biggest boons to humanity is water's presence on 70% of the earth's surface, primarily used for survival and sustenance and secondarily for travel, transportation, and generation of revenues. However, marine biological species' footprints leading to fouling are one of the biggest challenges in the maritime sector, be it for the global

shipping industry or sea/freshwater cooling installations or offshore marine structures, or scientific sensor installations (Gore et al. 2019a, b, 2020b, a, 2021; Pawar et al. 2019; Korde et al. 2020; Issac and Kandasubramanian 2021). An estimated annual approximate expenditure of 5.7 billion US dollars and 277 million US dollars was expended specifically to regulate oceanic biological foulants both in saline and freshwater atmospheres, correspondingly (Rouhi 1998; Nakano and Strayer 2014). Obscuration of offshore marine assemblies (resultant in prolonged inspection interval), conception of microenvironments promoting deterioration, bodily hindrance, augmented weightiness, and altered hydrodynamic profiling are general perceptible outcomes of biological fouling. All of these factors have considerable budget repercussions concerning the inspection, preservation, and overhaul of offshore installations. The term fouling refers to an undesired filthy growth on material (inner or outer or both) surfaces, either by living organisms or by non-living substances. Biofouling indicates colonization by living organisms, including their by-products on material surfaces present in the oceanic atmosphere resultant in the detriment

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of desired functions (Callow and Callow 2002). Extensive research on the identification of biofoulants has discovered over 4000 species circa, but a minuscule of the entirety till date (Thiyagarajan et al. 1999; Yebra et al. 2004).

The foulants are broadly classified as micro and macro foulants (Fig. 1). Microfoulants include diatoms, bacteria, algae spores, while macrofoulants include barnacles, tube-worms, bryozoans, mussels, hydroids, tunicates, spirorbis (Phang et al. 2008; Li et al. 2010).

Brownish tintured one-celled algae encircled in a periphery composed of silica are known as diatoms that adhere to any favorable surface by the discharge of mucilaginous extracellular polymeric substances abbreviated EPS. This mainly constitutes polysaccharides abling diatoms to have “gliding” motility. Further quick division of cells of adhered diatoms to form colonies and subsequently coalesced to form a compact biofilm of a thickness of 500 μm (Callow and Callow 2002).

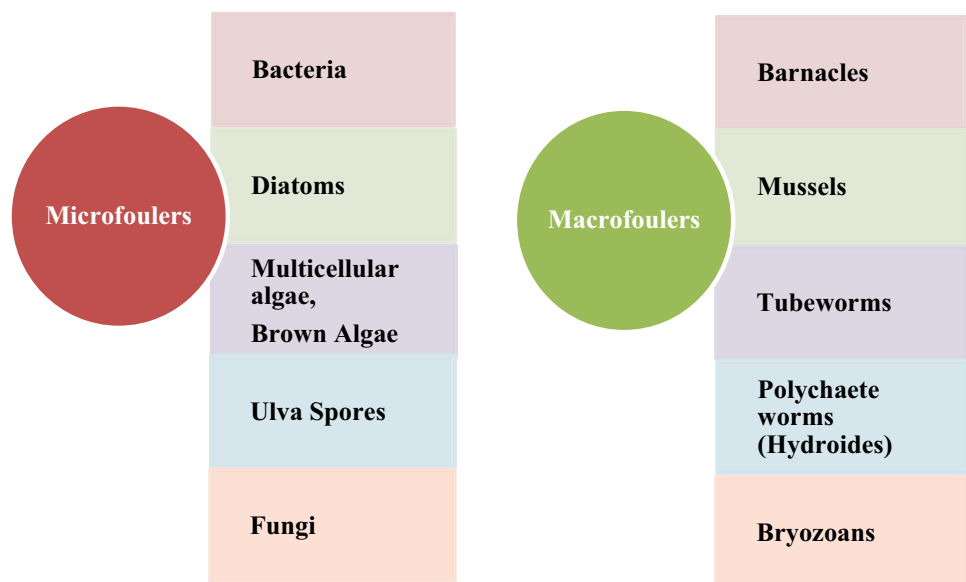
The colonization of *Ulva* which are a very small genus belonging to the family Ulvaceaea, on unsullied surfaces, occurs during the creation of minuscule motile zoospores (5–8 μm long). After the detection of a favorable substrate, swimming spores establish and get glued on the same by secretion of a hydrophilic glycoprotein bioadhesive. Noticeably, in few hours of adhesion, spore germination commences, i.e., cell division and growth, leading to sporelings (Lejars et al. 2012).

Marine biological fouling prohibits marine mobile and non-mobile structures from their designated service or functions and increases the operational cost (mainly fuel consumption and maintenance expenditure) (Sarkar and Kandasubramanian 2021b, a). The revelation of biofouling is through colonization of vessels, sonar instruments, floating

bases, buoys, oil rigs, offshore constructions, submerged cables, and seawater-cooling equipment. While designing marine vessels and offshore platforms (both commercial and military standards) for optimum hydrodynamic efficiency, a naval architect performs resistance and powering analysis for the economical and viable operation of the designed structure. To overcome the total resistance (residuary resistance and frictional resistance), equivalent powering calculations (shaft power) are thereby finalized. Biofouling alters the designed hydrodynamic flow stability of the vessel, escalates the surface roughness of the naked hull or the submerged truss structure of offshore marine platforms resulting in an escalation of the frictional resistance, consumption of fuel, and deviations from the prescribed operational performance. Generally, fouling is observed around the submerged hull area, propeller (blades and hub), rudders and struts, and sea-inlets. The level of fouling varies with the type and condition of the coating existing around the hull, the region of deployment of the vessel, and the operational regime (Inamdar et al. 2018; Bangar et al. 2018b, a; Jayalakshmi et al. 2019). The upsurge in frictional resistance and operational and maintenance cost is tremendous due to calcareous hard foulants (e.g., oysters and barnacles) against soft foulants (e.g., bacterial, and diatomaceous slime and algae).

A schematic representation of the components contributing to a seagoing vessel’s annual operational expenditure and the consequential effects of biofouling are highlighted in Figs. 2 and 3, respectively. Biofouling adds on the extra cost to the sub-components like fuel consumption (increased fouling leads to an upsurge in resistance, impacting augmented fuel consumption) and hull scrubbing (increased fouling lead to an increase in hull cleaning activity, thereby increased drydock period and charges along with extra labor

Fig. 1 Classification of biofoulants



cost) (Schultz and Swain 2000; Schultz 2004, 2007; Schultz et al. 2011).

Biofouling has also contributed to the unwanted invasion of alien species in marine habitats. The invasion occurs by the introduction of nonnative or non-indigenous species in new geographical locations either by adhesion to the external ship hull (as fouling organisms) or internally in ship ballast tanks (loading and unloading of ballast water). This unwanted invasive species leads to the reduction of biodiversity, threatens the existing species, degrades water quality

and habitats, reduces productivity in resource sectors such as fisheries and aquaculture, and, last but not least, most importantly, introduces diseases (Davidson et al. 2009).

A study conducted on full-scale resistance and powering with a series of roughness and fouling criteria has highlighted the following (Schultz et al. 2011):

- a. At cruising speed criteria or normal running conditions, it predicted that the total resistance increased drastically from 0% for a smooth, hydrodynamic hull to 11% for a

Fig. 2 Bifurcation of annual operational budget

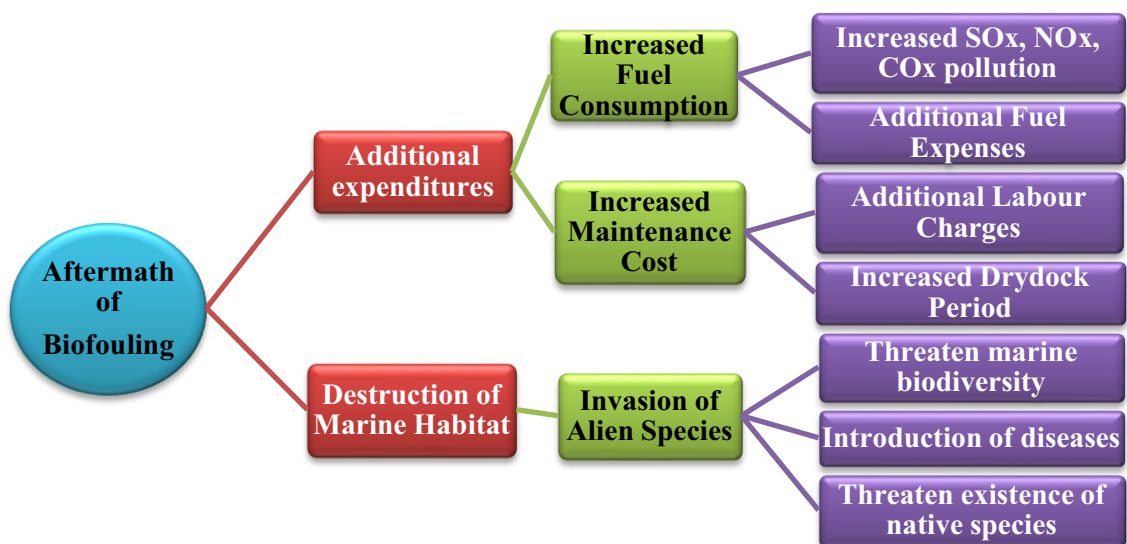
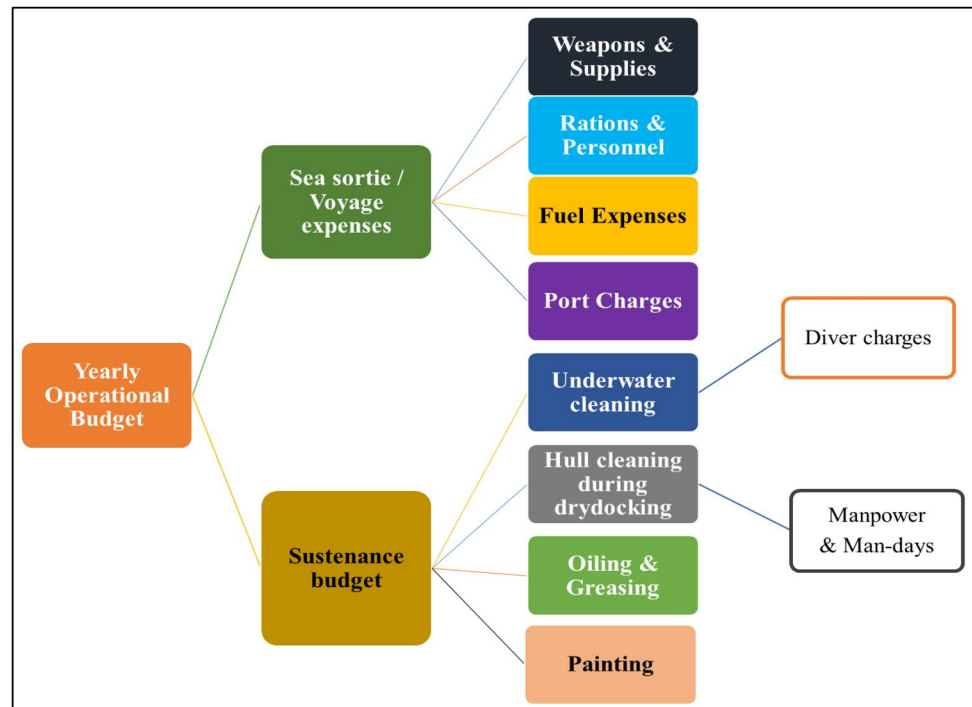


Fig. 3 Domino effects of biofouling

deteriorated antifouling coating or light slime layer and finally 80% for a substantial layer of hard shell foulants. Similarly, the requisite shaft power varied from 0 to 86% for above similar fouling conditions.

- b. At full speed criteria or mission-oriented condition, it predicted that the total resistance increased drastically from 0% for a smooth, hydrodynamic hull to 10% for a deteriorated antifouling coating or light slime layer and finally 55% for a substantial layer of hard shell foulants. Similarly, the requisite shaft power varied from 0 to 59% for above similar fouling conditions.

The estimated annual expenditure of US Navy to mitigate biofouling on ship's hulls was US\$56 million. This expenditure was inclusive of the cost of scrubbing the exterior of hull surface and reapplication of new coats of paints; nevertheless, the bulk of the expenditure primarily comprised of augmented fuel consumption contributed by biofouling (Schultz et al. 2011).

The establishment of surfaces by biofoulers and the subsequent humungous expenses associated with transportation and delivery delays, upkeep of vessel's hull, scrubbing of desalination systems, and biocorrosion (with an annual estimate of \$150 billion) paved the way for innovating typical coatings that should be resistant to fouling (Maréchal and Hellio 2009). In the 60 s, capable fouling-resistant paints consisting of organotin combinations like tributyltin (TBT) and triphenyltin (TPT) were researched and developed by the chemical industries. However, these compounds were exceptionally toxic to abundant maritime creatures and also highlighted their indomitable presence in the ecosystem. TBT concentrations nearby to 1 n/gI highlighted noteworthy modifications in marine gastropods, crustaceans, whales, and dolphins (Alzieu 2000). Following the prohibition of tributyltin comprised paints, the development of new formulations comprising higher copper and herbicides were noticeable. These paints have claimed to be ecologically friendly during their debut for commercial applications, but evidence of significant concentrations of these compounds in many countries (Europe, North America and Japan) was awakening (Turner et al. 2009; Gharde and Kandasubramanian 2018; Thakur et al. 2019). Besides, it was revealed that bacteria existing, especially in estuaries, when came in contact with these fouling-resistant lethal paints, had rapidly developed resistance to these biocides (Kobori et al. 1984; Mudryk 2002). The commercial applicability of these coatings should be effectual in such a manner that drydocking of vessels could be reduced in a significant proportion by permitting at least 5 years of fouling resistance. It should be resilient and resistant to mutilation, has lesser maintenance, ease of reparability, and ease of application. The coating should ensure smooth hydrodynamic profiling as well as compatibility with existing coatings. Criteria like

cost-effectiveness, harmlessness to non-targeted organisms, and effective functionality both at the ocean and port (Ralston and Swain 2009) would promote these coatings industrially.

This review highlights the study of the lifecycle of barnacles, the process of barnacle fouling, various advances on the development of environmental-friendly coatings (both natural and synthetic additives) used for suppression of barnacle establishment till date, general life cycle assessment, and future perspectives in this field of anti-barnacle coatings. The areas of evolution for the need of coatings, their performance against foulants, and the protection tenure on the applied surfaces, various alternative environmental-friendly physical and chemical methods of mitigating barnacle establishment have also been focused upon and squeezed into this review for appreciation. When evaluating new preparations or compounds for prospective fouling resistance applications, limitations like periodic obtainability may confine maximum biological researches to relatively few taxa and species. Several approaches like application of protective coatings, material alteration, surface topography modification, alteration of environmental factors, and ecologically safe techniques have been researched and investigated; however, conversion into commercial applications is still a far-fetched reality.

Evident manifestations of biofouling on marine vessels

The exterior form of the vessel's body is of principal prominence for the operation of seafaring crafts. Unwanted biofouling on ship hulls escalates the surface roughness leading to loss of smooth hydrodynamic water flow, thereby amplifying the frictional resistance and effective powering of the vessel. Hence, commercial operators have to be decisive either to raise the power component to maintain designated maneuvering speed, which amplifies the consumption fuel oil and also emit harmful greenhouse gases or to decrease the speed at fixed shaft power thereby inviting a delay in the arrival time to the destination which would definitely lead to contractual, monetary consequences. Hence, scrubbing of foulants becomes a priority either to mitigate the loss of the vessel's operational characteristics or to undergo intermittent hull cleaning at drydock or in afloat condition. The two prime evident impacts of biofouling are enumerated below:

- (a) Alteration of designed vessel hydrodynamics characteristics:

After the initial assessment of the effect of roughness on the hull of a destroyer class military vessel conducted by William Froude in 1872 and 1874, numerous experimental studies since the past 140 years had been dedicated to

analyze the consequence of hull roughness factor on skin friction parameter. There has been abundant research investigating the adversarial consequence of naval specific coats and biological fouling on vessel's resistance by piloting innumerable investigational approaches, including the towing test of flat plates and rotating disc flow channel. The most broadly applied technique to envisage the effect of roughness on a full-scale ship resistance is the scaling of the boundary layer similarity law proposed by Granville in the years 1958 and 1978. It is important to understand that vessel hydrodynamics, vessel resistance, and powering are interrelated parameters. A deviation from the designed hydrodynamic profile augments the resistance component thereby affecting the designated shaft output of the vessel. (Song et al. 2019).

Y. K. Demirel et al. (2017) had executed an array of towed assessments with the help of flat plates covered with synthetic barnacles in a towing tank. These barnacles were manufactured by the help of 3D printing and were segregated into various dimensions to improvise the actual scenario of fouling of immersed hull. The synthetic foulants were bonded against the exterior of the monoplane foils fabricated from of stainless steel 304 grade and then in a towing tank at a various velocity, towing tests of these plates were conducted. These assessments were intended to investigate the impact of height of barnacles and their coverage percentage on parameters like the vessel's resistance and effective power. The evaluation of parameters like coefficient of drag and roughness function was also assessed. The prediction of the impact of roughness by the foulants on the vessels' frictional resistances was undertaken, which helped to plot the added resistance diagrams. Finally, actual consequence on powering parameter of the vessels was calculated. The outcomes specified the influence of dimensions of barnacle was substantial. The coverage by 10% of barnacles of height 5 mm each resulted in analogous requirements of additional power due to coverage by 50% of barnacles of height of 1.25 mm each. The vital parameters like augmented resistance coefficient and augmented effective power due to surface roughness displayed a growing positive trend with rise in surface coverage area by barnacles.

D. Uzun et al. (2020) also performed an array of towing tests using flat rectangular profiles fabricated from stainless steel 304 grade with analytically covered 3D-printed barnacle tiles. The tests investigated the consequence of establishment of barnacles on oceanic vessels parameters like its resistance and effective power. The difference between the towing test experimental setup by Y. K. Demirel et al. (2017) and D. Uzun et al. (2020) was that the latter had implemented an additional surrounding intended to represent tangible natural establishment of barnacle which is called a chaotic settlement. Moreover, the dimensions of barnacles were more varied in investigations performed by D.

Uzun et al. (2020) than in investigations performed by Y. K. Demirel et al. (2017). An all-encompassing evaluation was drawn among this natural and designed settlement in accordance with the standards. By applying the law of boundary layer similarity the parameters like coefficients of drag and values of roughness functions were assessed and full-scale ship resistance and powering were assessed for six diverse ships at their cruising speeds. Additionally, at the designated operational power, the decrease in the cruising speed due to fouling of barnacles was also assessed. The outcome of the investigation was that at cruise speeds regime, the settling pattern resulted in 10.5% variance in frictional resistance and 6.7% variance in powering and at fixed effective power the settling pattern resulted in 20.5% reduction of velocity.

Osman Turan et al. (2016) had also performed an array of towing tests using flat plates fabricated from of stainless steel 304 grade covered with artificial barnacles. These investigations were intended to scrutinize the consequence of the percentage exposure of barnacles on the component of resistance and component of operational power of crafts as a function of Reynolds numbers. These tests highlighted the consequences of hard foulants for various speed regimes (Reynolds numbers) on the added resistances concerning the coefficient of added frictional resistance for an exterior coverage proportion of 20% of fouling. An in-house code was solely developed for evaluation of the drag coefficients and surface roughness function values. These generated codes were centered on the scaling method of the similarity law elucidated by Schultz (2007) for a specified situation of fouling conditions. These codes also considered the assessment of the similarity law of Granville. The experimental analysis revealed that with the increase of fouling's surface coverage, the frictional resistance and effective power also displays a steep rising trend. This methodology assumed a standardized scattering of foulants on the monoplane foils, which may differ from real life scenario.

The three latest researches performed by Y. K. Demirel et al. (2017), D. Uzun et al. (2020), and Osman Turan et al. (2016) revealed an analogous underlying phenomenon, that the macrofoulants on the immersed hull surface altered the designed hydrodynamic profiling of the outer skin of the vessels causing transformation of water flow from laminar to turbulent regime. The growth of macrofoulants also adds to the surface roughness, increases skin drag, and augments the vessels' mass. All these have a cascading effect on the vessel's total resistance, thereby amplifying the effective shaft output.

(b) Hull cleaning and maintenance regime and associated expenditure:

M.P. Schultz et al. (2011) analyzed the complete financial effect of fouling of the exterior surface on a naval ship of

medium size. An array of overheads allied with surface fouling was scrutinized, comprising cash outflows for fuel consumption, coatings of exterior, application and removal of hull coating, and hull cleaning. The inclusive budget linked with surface soiling for the Navy's existing coating, scrubbing, and extent of soiling was estimated to be US\$56 million per year for the designated class of warship which sums up to \$1 billion for 15 years. This study also revealed that the expenses concerning the scrubbing of hull and reapplication of paints were considerably lesser than the expenses concerning fuel. Swain et al. (Swain et al. 2007) compared the obtained values of frictional drag obtained during field experimentations and revealed that the operational parameters of both the coatings of fouling resilient and fouling release nature when deployed in mobile and immobile conditions vary significantly from each other. Under dynamic conditions, the surfaces of fouling release nature revealed much lesser frictional drag in comparison to static circumstances, and particularly these surfaces also revealed much lesser frictional drag in comparison to copper-centered fouling resistant coatings. The Naval Ships' Technical Manual (NSTM) (2006) dedicated for US Navy issues directives for the conduct of inspections of immersed hull for presence of fouling and coating damage in a systematic manner. The verdict on whether to scrub or hold is based on the outcomes of consistent inspections and is not specified by the US Navy. An example from NSTM 2006 is that whole hull cleaning is considered when fouling proportion covers 20% of the hull (for ablative fouling resistant paints) or 10% of the hull (for fouling-release coatings). The Naval Ships' Technical Manual version 2006 highlights various methods of afloat conditions of cleaning to restore performance. During the entire cleaning procedure, the fouling is scrubbed from the complete submerged hull and appendages like propellers, shafts, struts and rudders, and all openings. In an interim cleaning procedure, fouling from propellers, shafts, struts, and rudders are exclusively removed. In a partial cleaning procedure, the fouling from precise segments of the ship hull is removed, and this can also be accomplished along with an interim cleaning. The investigation period revealed that the typical occurrence of full hull cleanings was 0.21 per annum, whereas the typical occurrence of interim cleanings was 2.4 per annum. A full cleaning expense ranged from \$26,200 to \$34,200 and for an interim cleaning ranged from \$15,000 to \$21,500. Every 7.5 years, whole hull repainting was carried out. The painting's expenses are inclusive of the budget rate of the paint that has been consumed and workforce for surface readiness and application. The comprehensive hull coating process was carried out during the drydocking phase and the model presumed that the other maintenance apart from patch up was also executed simultaneously at the drydock interval, and these upkeep activities absorbed all the drydocking expenses. The bottom-line of the research was a

reduction in the fouling from Fouling Rating-30 to Fouling Rating-20 resulted in a yearly budgetary profit of \$340 K per vessel to the Navy, or an overall of \$19 M for the entire specific class of vessel. A schematic overview of biofouling prone locations on a seafaring vessel is highlighted in Fig. 4 and a schematic pie-chart visualization of the economic impact due to biofouling is depicted in Fig. 5.

Stages of biofouling

The sessile forms of micro and macro-organisms are mainly responsible for biofouling on any immersed suitable substrate. The stages of biofouling are subdivided into four discrete phases (Fig. 6), namely:

- (a) Initiation of formation of a biofilm layer, known as conditional film is the foremost stage. This is formed by primary adhesion of physically adsorbed organic molecules (proteins, proteoglycans, polysaccharides, glycoproteins). Biofilm or slime formation takes a few seconds to minutes. The process may be reversible; that is biofilm may be washed away under external circumstances.
- (b) Primary accumulation or colonization by bacteria and single-cell diatoms and their secretions on this altered surface leads to an increase in the biofilm thickness and also the adhesion strength. Primary colonization formation takes 1 h to 1 day's duration for permanent adhesion on the substrate.
- (c) Secondary accumulation or colonization is by capturing microorganisms (spores, protozoa) and settling on the primary accumulation. Secondary colonization formation takes a minimum of a week.
- (d) Tertiary accumulation or colonization is by increased capture of microorganisms as well as macroorganisms (sponges, polychaetes, mollusks, macroalgae barnacles, bryozoans, and tunicates). Tertiary colonization formation takes a minimum of a month.

The formation of the biofilm or the slimy layer is the general indication for the onset of biofouling. However, studies have shown that many species (like zoospores of the alga *Ulva linza* and cyprids of the barnacle) can even establish on unsullied substrates, indicating that presence of conditioning film is immaterial. (Lejars et al. 2012; Dang and Lovell 2015)

Main macrofoulants—barnacles

Barnacles (*Balanus glandula*) are gluey crustaceans like crabs, lobsters, and shrimps. Adult barnacles are sessile and are covered with calcareous plates and are cemented in

Fig. 4 Schematic overview of locations prone to biofouling of a seagoing vessel

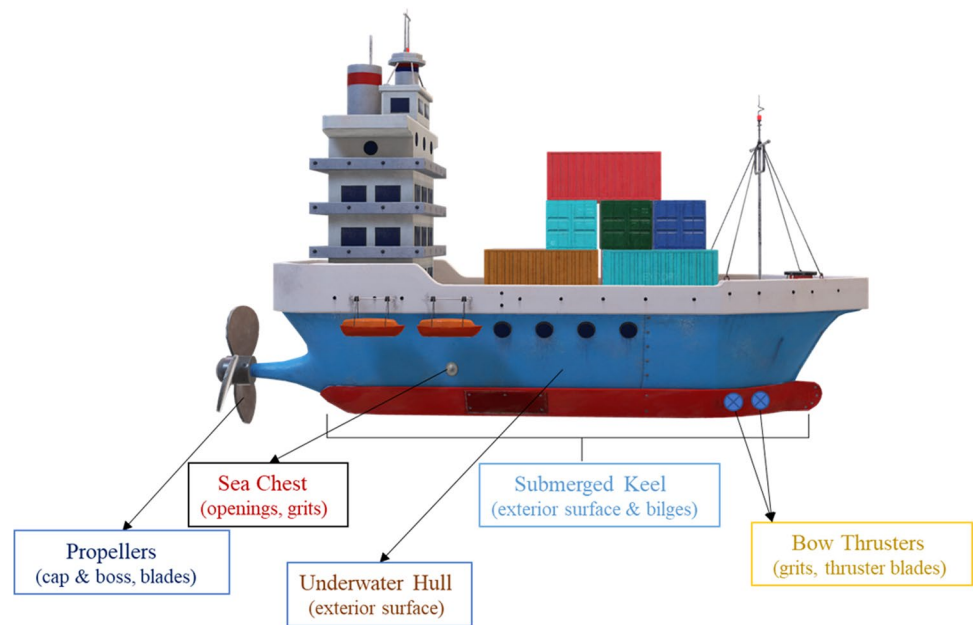
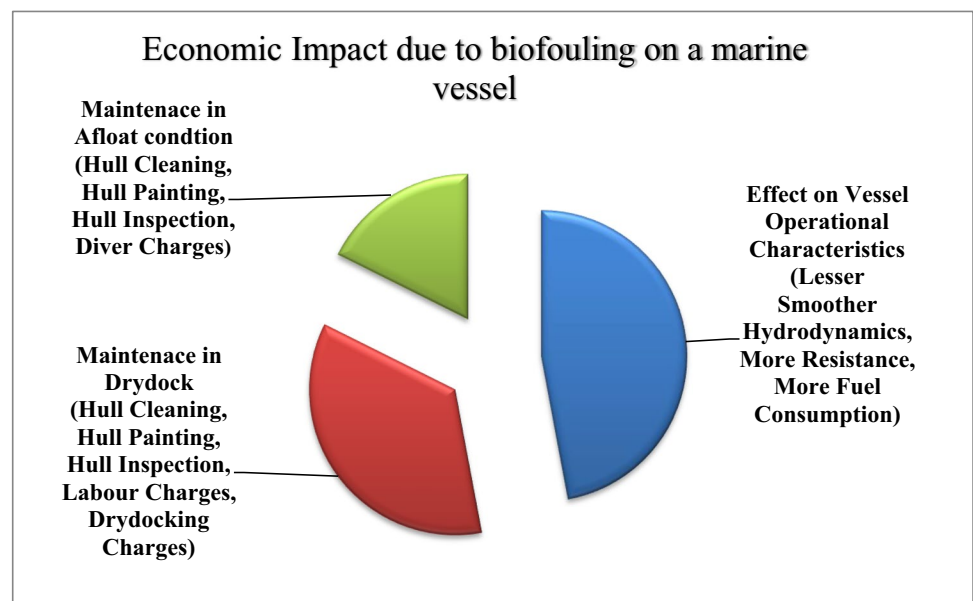


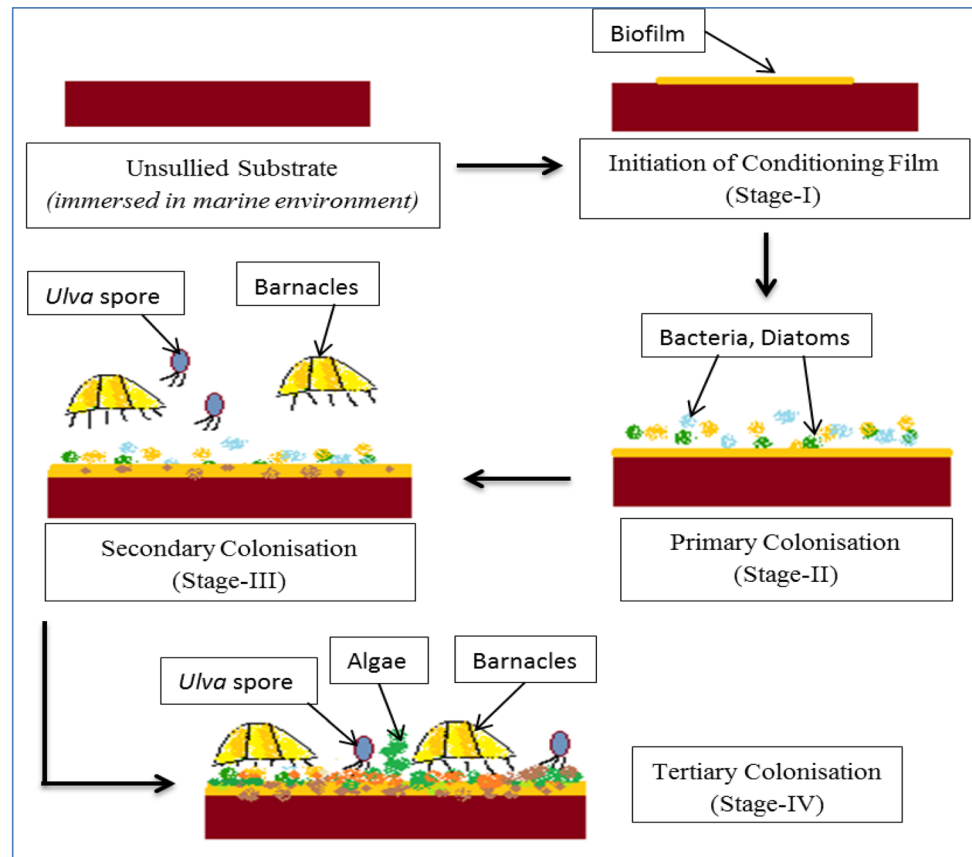
Fig. 5 Economic effect due to biofouling on a marine vessel



an inverted manner (i.e., forehead down) to substrates like rocks, marine mobile, and immobile platforms, the outer skin of aquatic mammals. Barnacles have naturally adapted feathery retractile organs known as cirri, which help them to sustain by capturing minuscule food particles for sustenance by performing a combing motion. Hence, barnacles are also recognized as Cirripede (National Oceanic and Atmospheric Administration 2020; Maréchal and Hellio 2011).

The stages of growth of barnacles

The growth stages of benthic marine invertebrates involve motile pelagic larval phases and sessile adult phases. The planktonic larvae' duration in the water column varies from minutes to months, depending upon their surroundings and favorable surface for establishment. The barnacle (*Balanus amphitrite*) has planktonic naupliar phases of six in numbers, tailed by the cyprid phase. Successful attachment on the substrate is the key for the larva to finally metamorphosis into a sessile benthic adult barnacle. The major four stages

Fig. 6 Stages of biofouling

of a barnacle life cycle are depicted in (Fig. 7) (Rajitha et al. 2020). In the initial days of a barnacles' lifecycle, the nauplius larvae stage feeds on plankton. The final larval stage is the cypris (approximately 500 μm long) has unrestricted movement in water before settling on a surface. For the final transition from a nauplius larvae stage to calcified adult barnacle, these cyprids explore and finally adheres to the rigid substrate. In the course of this exploration phase, the cyprid "walks" on the exterior by means of a pair of body part accessory or "antennules" that discharges an adhesive of impermanent in nature. This exploration phase is a reversible process, denoting that during the process of exploration by cyprids, they can detach itself from the surface, leaving behind a rounded outgrowth of the temporary adhesive as markings. These markings are resistant to biodegradation as well as cannot be disseminated in aquatic medium and thereby functions as a signaling or communicating molecule for further establishment of additional cyprids, subsequently metamorphoses into the calcified adult barnacle (Lejars et al. 2012).

Essential steps to barnacle's establishment

A barnacle cypris larva, the prior onset of its juvenile phase, swims in the water column with the help of its six pairs of

**Fig. 7** Growth stages of a barnacle. Reprinted with permission from (Rajitha et al. 2020). Copyright (2020), ELSEVIER

thoracic legs. It is observed that the chemosensory lattice body part located on the carapace of larvae steer the larvae in the path which is favorable for establishment (Bielecki

et al. 2009). Cypris consists of a couple of frontal aligned antennules with six pairs of thoracopods. The first pair of antennae of the nauplius is adapted as a sensory tool that helps in the navigation of cypris in search of a suitable substrate for establishment. The first two segments of the antennules of cypris help in mobility and the remaining are equipped with sensory establishment functions amounting to a total of four segments. They have multiple organs of chemo-mechanoreceptor and are sustained by the nervous system. During the pre-establishment exploration, the indications, i.e., the cues emerging from the marked sites, come in contact with these receptors resulting in sensing the site's potentiality. The third segment is equipped with cuticular villi covered adhesion organs (antennular disc). During the survey phase, these structures aid in momentary connection by holding the cyprid to the surface. The chemical hints originating from conspecific adults, target organisms, or surfaces of substrates are also crucial to barnacle establishment. The key step in the lifecycle of a barnacle process (Fig. 8) is reliable and permanent adhesion that leads to the successful attachment of a cyprid on the substrate (Anil et al. 2010).

Factors affecting marine biofouling

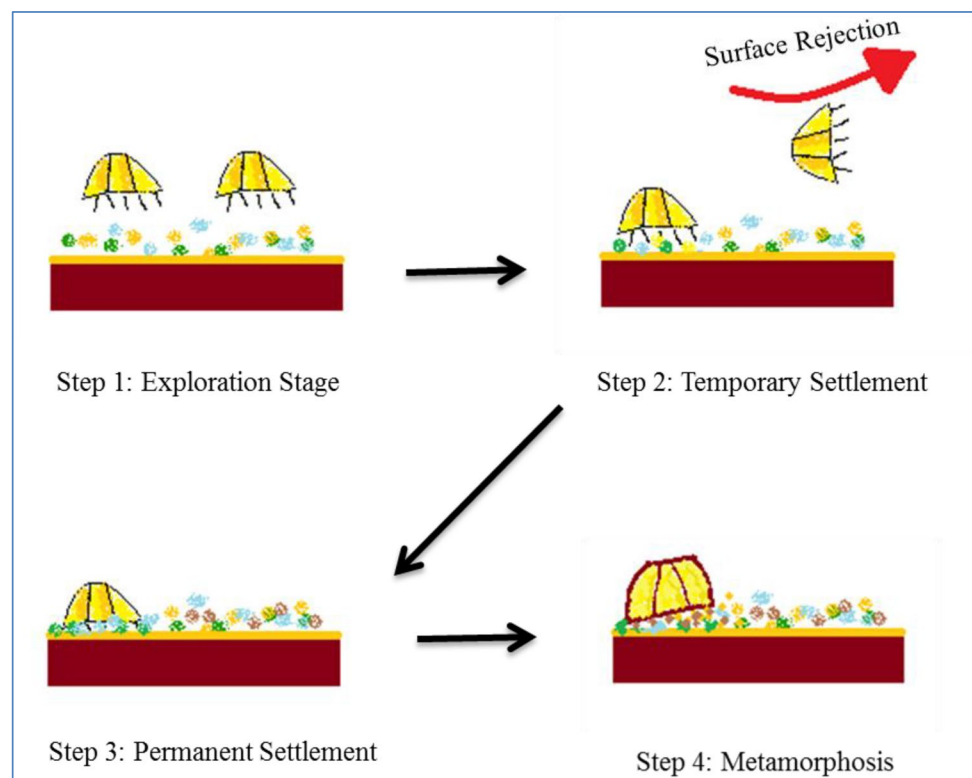
The environmental parameters, namely salinity, pH, temperature, density, nutrient levels, underwater current rate, exposure and amount of solar radiation, the concentration

of dissolved gases, and topology of the substrate affect the accomplishment of colonization by the foulants. These factors are geographical locations, seasonal changes, and alteration of water depth-dependent. However, salinity, pH, temperature, and density majorly contribute to marine biofouling's growth rate (Fig. 9.). In Polar regions, colonization occurs only during the summer season, which lasts for a few weeks and negligible occurrence during the rest of the season. Dissolved gases have significant attributions to the level of corrosion and biotic growth in the marine environment (Palaniappan et al. 2019, 2020; Korde et al. 2020; Cole et al. 2021). It is believed that the upper layer of the ocean is soaked with aerial gases (the majority being nitrogen, oxygen, and carbon dioxide); however, various biological processes like breathing and photosynthesis can modify these concentrations. Any algal action can supersaturate the upper layers of the ocean surface (Yebra et al. 2004).

Salinity

The salt content of seawater, distant from near coast effects like melting of ice, freshwater of rivers, and parts of greater evaporation, ranges between 3.3 and 3.8 wt.%. Higher contrast of the salinity content near the water surface is due to the amount of rainfall and evaporation experienced in that region. Generally, higher salinity content is observed in the tropics than the equator zone. The universally accepted salinity of seawater is 3.5 wt.%, although drastic variations

Fig. 8 Steps of barnacle establishment



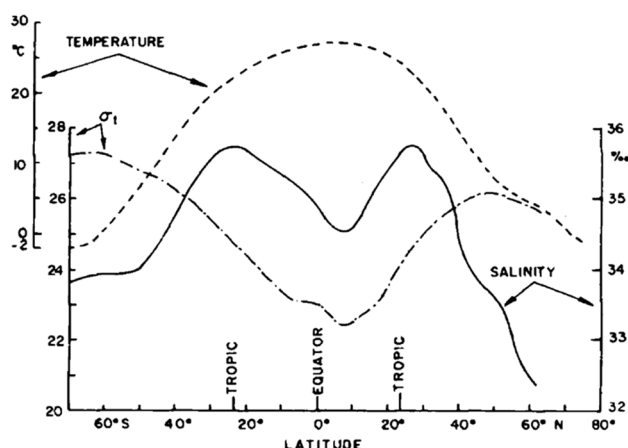


Fig. 9 Variation of environmental factors with latitude affecting biofouling process. Reprinted with permission from (Yebra et al. 2004). Copyright (2004), ELSEVIER

at the poles are observed. Lower salinity is unfavorable to growth rate, attainment to maximum size, and also it leads to several malformations of the foulants (Yebra et al. 2004).

Temperature

The temperature of seawater ranges amid -2°C to 28°C at the poles and the Equator, respectively, locally recording a temperature of 35°C . The temperature of the hydrosphere is less affected by external weather conditions in comparison to the lithosphere. The diurnal variations of temperature of ocean surface hover around 0.4°C . However, annual variation of temperature of oceans surface is noticeable due to varied solar radiation exposure and absorption, evaporation from the ocean surface, rainfall, and lastly, the amount of heat interchange with the atmosphere (Yebra et al. 2004).

pH

The marine environment is naturally alkaline, and the ocean surface pH ranges from 8.0 to 8.3. The carbonate system (CO_2 , HCO_3^- , CO_3^{2-}) offers a buffer capacity to seawater. In an environment with numerous microbiological activities, expected variations in pH are due to the hydrogen sulfide production (reducing of pH) or scavenging of CO_2 by algae (increase of pH). Other factors like temperature also influence the pH value; a surge in temperature drops the pH value unless excessive desorption of CO_2 escalates the pH value (Yebra et al. 2004).

Surface topology of substrate

The substrate's surface energy has a substantial consequence on the encampment and the bonding power of foulants.

Three generalized patterns concerning surface energy of substrate have been outlaid which are the following:

- The encampment of bryozoans on the substrate with surface energy ranged between (10 and 30 mN m^{-1}), indicating that they prefer low surface energy material for establishment (Rittschof and Costlow 1989).
- The encampment of barnacles on the substrate with surface energy ranged between (30 – 35 mN m^{-1}), indicating that they prefer high surface energy material for establishment (Rittschof and Costlow 1989).
- Hydroids' encampments are irrespective of the surface energy of substrates.

Interestingly, it was observed that increase or decrease of bacterial adhesion was affected with an increase in substrates' surface energy for establishment, along with several other factors like physical and chemical properties of bacteria, substrates (like electric charge and hydrophobicity), and the marine environment (Dexter et al. 1975; Dexter 1979; Fletcher and Marshall 1982; Ista et al. 2004; Liu and Zhao 2005). Roughness and porosity factor also influenced the encampment of fouling organisms on a submerged substrate because surface undulations led both to an augmentation of surface area and adhesion sites resulting in higher chances of colonization (Fletcher and Callow 1992; Howell and Behrends 2006). The troughs of coarse surfaces breached by marine bioadhesives created a secure mechanical lock on the substrate surface upon curing (Candries et al. 2001).

Evolution of biofouling—a brief insight

Commercial shipping being the prime facilitator of global trade and contributing major revenue to economic growth, led to exponential growth in the marine shipping industry. On the other side, this led to an increase in biofouling of marine structures resulting in augmented fuel consumption to substantiate the reduction of speed due to unwanted fouling, excessive emission of SO_x , NO_x , and CO_x , transportation, and introduction of alien species, and lastly, increased maintenance expenditure to maintain the marine vessels. A thoroughly detailed introspection revealed the documented detrimental consequences of fouling on the vessel's performance, and remedial measures to prevent fouling were already implemented since the seventh century BCE. Ancient Mediterranean civilizations such as Phoenicians, Carthaginians, Greeks, and Romans had already testified their novel attempts on antifouling (Kiil et al. 2006; Selim et al. 2017).

In the seventh century BCE, Phoenician galleys marked lead-sheathed timber debut as fouling protection followed by usage of arsenic and sulfur-and-oil

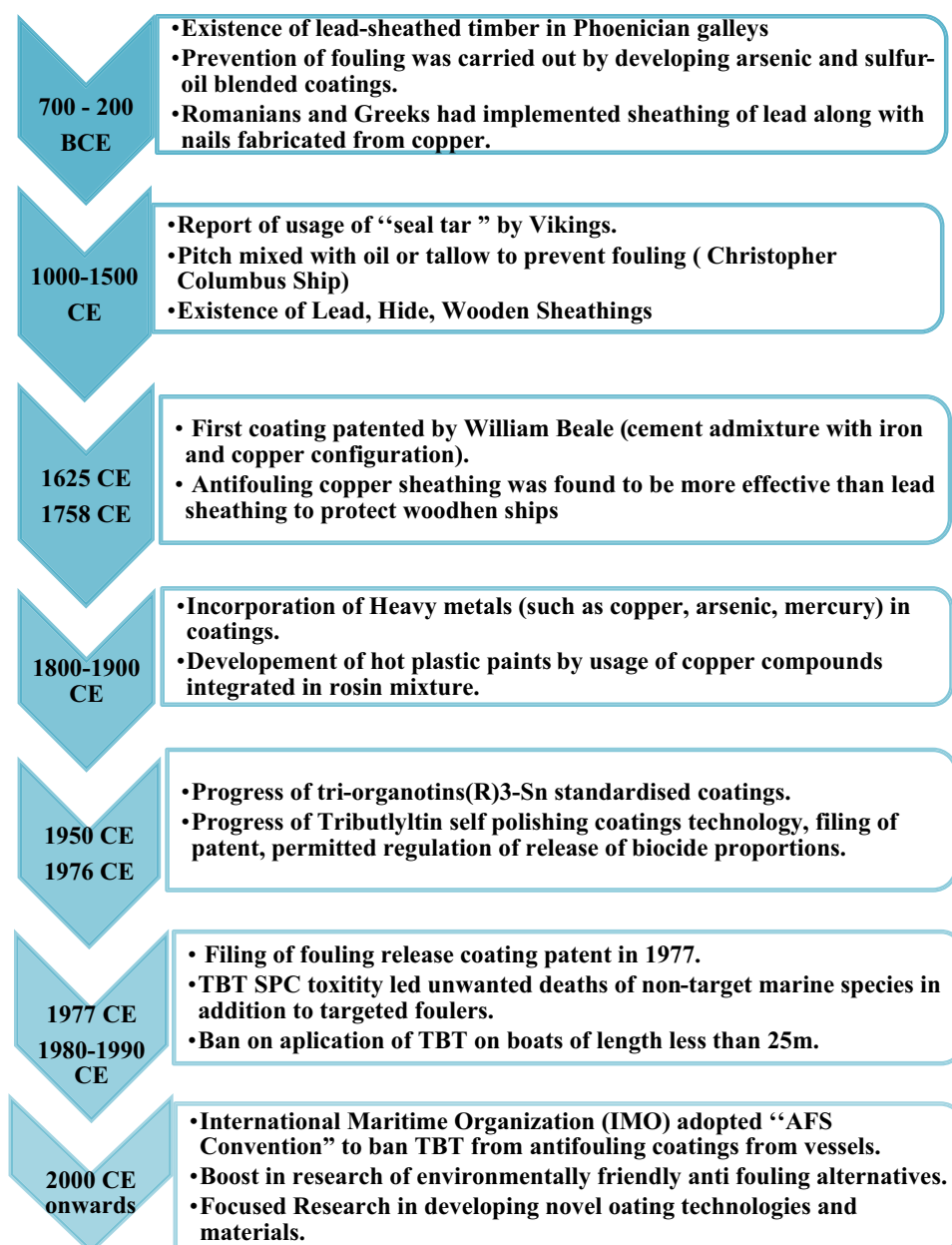
mixture coatings and lead sheathing secretions along with copper nails by Romans and Greeks for prevention of fouling in fifth century BCE and third century BCE, correspondingly (Kiil et al. 2006; Selim et al. 2017). The usage of “seal tar” was reported several centuries later by the Vikings. During thirteenth to fifteenth centuries, wide usage of pitch to avert biofouling was noted and blended with tallow or oils used during the expedition by Christopher Columbus. In the fourteenth century, application of hide sheathing was also documented. During sixteenth century, boats were enclosed with woods coated with layer of fat of animals. In the seventeenth century, the British’s banned the use of lead sheathings due to severe rusting of steel materials like ship rudders (Yebra et al. 2004).

In 1758, antifouling of copper sheathing was examined in Britain, and this analysis resulted in more resourceful outcomes in comparison to sheathing of lead. This discovery led to extensive utilization of the technology in the governmental fleet ever since inception and was the most desired methodology to shield hulls of timber-made boat. But with the arrival of ships fabricated from steel and subsequently the related corrosion issues allied with metallic enclosing suspended the application of this technology and focused on alternate antifouling strategies. In the middle of 1800, William John Hay had segregated iron fabricated hull from metallic toxicants by applying resins with major quantities of arsenic, copper oxide, and mercury oxide dispersed in it and by application of non-conducting varnishes (Dafforn et al. 2011). US Navy’s research study in the early twentieth century highlighted that rosin established beneficial properties, like being the abundantly available, cheaper, and successful replacement of rare and expensive shellac. For enhanced performance, blending of rosin with lethal compositions of mercury or copper was also carried out (Yebra et al. 2004). Various remarkable events during the progress of biofouling have been indicated in the schematic timeline (Fig. 10).

A tributyltin biocidal copolymer when comes in contact with aqueous medium then poisonous substances are released in the oceanic atmosphere. Primarily, tributyltin combinations were primarily employed in advanced copper antifouling coatings and were gradually implemented as an antifouling system in totality, which went through “free association form” within the matrix. Tributyltin coatings of self-cleaning and buffing nature comprise esters of tributyltin acrylate as antifouling coatings developed in the year 1958. In the year 1964, James patented a self-polishing coating based on copolymers of tributyltin acrylate and methyl methacrylate (Champ 2000; Qiu et al. 2008). In common, tributyltin–self-polishing coatings technology slowed the swift dissolution of tributyltin-bearing compounds which were incorporated

in free-associated form; thereby, a long durable foulant resistance was ensured. Tributyltin–self-polishing coatings is advantageous because it ensures constant biocide release rates over time, smooth hydrodynamic profiling of the hull during navigation leading to optimum resistance condition as well as desired fuel consumption. Unfortunately, the slow rate of degradation of Tributyltin-bearing biocides makes it readily bioavailable to non-target species (Champ 2000; Sjollema et al. 2014). Tributyltin has attacked non aimed aquatic creatures such as oysters and mussels and caused abnormal development of shell, brittleness of shells, and pitiable weight gain (Hallers-tjabbes and Boon n.d.; Santos et al. 2002). Further accumulation of tributyltin was found in various mammals like fish, dolphins, seals, whales. It also affected a varied range of invertebrates, causing sterility and even reported to be fatal. This ecological harm resulted in indirect effects of commercial fisheries and tourism (Hallers-tjabbes and Boon n.d.; Strand and Jacobsen 2005; Kotrikla 2009). The middle of the 1980s marked forbidden use of tributyltin on small boats (length of vessel should be less than 25 m) in numerous republics as an outcome of the ecological harm. An International Convention hosted by IMO on October 5, 2001 debarred the usage of tributyltin-oriented antifouling paints scheduled from January 1, 2003, and a total ban of the occurrence of these paints on hulls of ship were imposed from January 1, 2008 (International Maritime Organization 2008; Okoro et al. 2011). Though development of copper centered tin-free antifouling paints were in progress, the practice of the same has been under investigation in numerous nations and even subjected to environment regulations in terms of release rates of copper. Regardless of the greater level of copper’s lethal nature to abundant oceanic populations, common algae were comparatively more tolerant and could adhere to the substrate easily. The addition of booster biocides to copper antifouling coatings shadowed these problems and the paint community was advised to develop tributyltin-free products. Some less poisonous materials with lesser molar mass (organic biocides like Zinc pyrithione, Sea-Nine 211™, Irgarol 1051, Zineb, Diuron) integrated in the matrix had already been used (Maréchal and Hellio 2009; Thomas and Brooks 2010; Dafforn et al. 2011; Lejars et al. 2012; Schultz et al. 2011). The aggravated poisoning effect of these booster toxicants even at minor proportions and their non-biodegradability in aquatic environment have led to enforcement of regulations in few nations to control their usage (Blanck et al. 2009; Sapozhnikova et al. 2013). The washing out of antifouling coatings on the vessel’s outer skin has also caused damages to the exterior, like corrosion and staining, total oozing of elements, and shorter period of intermittent dry-docking. Various studies have been carried out to

Fig. 10 Timeline of the progress of antifouling coating. AFS Convention: The Convention on the Control of Harmful Antifouling Systems on Ships (AFS Convention), 2001



cultivate new non-biocidal coatings and non-sticky fouling release coatings with the mounting apprehension on the detrimental adverse effects of biocidal coatings on the surroundings (Yebra et al. 2004; Tsoukatou et al. 2007). The development of fluoropolymer and silicon-oriented non-sticky, fouling release coatings was also in progress. Instead of preventing the fouling organisms’ adherence on the exterior hull, the reduction of the strength of affixation of these foulants already present would lead to easy and swift removal during the motion of the vessel (Yebra et al. 2004; Lejars et al. 2012). The development of capable alternative was to intermingle covalently the biocidal role on a polymer chain to evade the discharge of biocide products, hence preserving the perpetual antifouling

action. Some biocides like salts of quaternary ammonia, salts of phosphorous, salts of sulfur, derivatives of chlorophenyl and *N*-halamine had been intermingled into conventional polymers. In paints with antifouling properties, quaternary ammonium salts are principally originated in polymers like polyurethanes (Nurdin et al. 1993a, b; Grapski and Cooper 2001; Kébir et al. 2007), polysiloxane (Gottenbos et al. 2002; Majumdar et al. 2008; Mizerska et al. 2009; Lejars et al. 2012), and polymethacrylates (Ravikumar et al. 2006; Kenawy et al. 2007). The need to substitute booster biocides compounds with ecologically benign compounds that are also barnacle resistant is the present necessity and elaborated and categorized in subsequent paragraphs.

Natural compounds

When exploring novel non-toxic antifoulants and their subsequent development into an antifouling product, the indicator to assess the effectiveness of an antifouling composition with respect to its harmfulness is termed as therapeutic ratio (LC_{50}/EC_{50}) where the toxic concentration or dosage of an antifouling composition for 50% of a fouling organism (e.g., barnacles, bryozoans) larvae is denoted as LC_{50} , while the effective dosage for preventing establishment and fatal to 50% of larvae of the fouling organism in comparison with the control is EC_{50} . The antifouling activities are expressed as the median lethal dose. Commonly, a composition with the ratio LC_{50}/EC_{50} greater than 15 is denoted as a non-toxic antifouling composition, but compositions with greater therapeutic ratios are desired. It has been observed that only small molecules with ratio of LC_{50}/EC_{50} greater than 50 and EC_{50} valued lesser than 5 $\mu\text{g/mL}$ are effective against both micro and macro foulants. The measurement of LC_{50} , EC_{50} along with the ratio of LC_{50}/EC_{50} for these compounds is reliant on the characteristics like stability and solubility. A comparatively lethal composition with lesser proportion of LC_{50}/EC_{50} can be prioritized for development and application provided that it is easily degradable (Qian et al. 2009; Rittschof et al. 2015).

Terpenoids and steroids

- (a) Avarol (sesquiterpenoid hydroquinone) (**1**) derived from *Dysidea avara* which is a Mediterranean sponge, exhibited ED_{50} measure of 0.65 $\mu\text{g/mL}$ and low toxicity LD_{50} measure of 13.3 $\mu\text{g/mL}$. It demonstrated to be an antifoulant to cyprids of the barnacle (*Balanus amphitrite*) (Tsoukatou et al. 2007).
- (b) Avarone (sesquiterpenoid hydroquinone) (**2**) derived from *Dysidea avara* which is a Mediterranean sponge, showed ED_{50} measure of 3.41 $\mu\text{g/mL}$, and low toxicity LD_{50} measure of 27.2 $\mu\text{g/mL}$. It also demonstrated to be an antifoulant to cyprids of the barnacle (*Balanus amphitrite*) (Tsoukatou et al. 2007).
- (c) 3'-(p-chlorophenyl) avarone (**3**) is one of the six prepared analogs of Avarone (**2**) which exhibited ED_{50} measure of 0.65 $\mu\text{g/mL}$ and LD_{50} measure greater than 50 $\mu\text{g/mL}$ in respect of cyprids (Qi et al. 2008b).
- (d) 4'-propylthioavarone (**4**) is one of the six prepared analogs of Avarone (**2**) which exhibited ED_{50} measure of 0.45 mg/mL and LD_{50} measure greater than 50 $\mu\text{g/mL}$ in respect of cyprids (Qi et al. 2008b).
- (e) Subergorgic acid (**5**) is sesquiterpene with a triquinane structure derived from gorgonian *Subergorgia suberosa* located in South China Sea. It subdued establishment of larvae of *Balanus amphitrite* (EC_{50} measure of 1.2 $\mu\text{g/}$

mL) and bryozoan *Bugula neritina* (EC_{50} measure of 3.2 $\mu\text{g/mL}$) (Qi et al. 2008b).

- (f) A primarene diterpene (**6**) derived from *Cerriops tagal*, a mangrove plant, showed encouraging fouling-resistant action concerning the barnacle *Balanus albicostatus* with ED_{50} measure of 0.04 $\mu\text{g/mL}$, and its noxiousness was on the lesser side (LD_{50} greater than 10 $\mu\text{g/cm}^2$) (Chen et al. 2008).
- (g) Renillafoulin (**7**) derived from a Caribbean sea pansy, out of the twelve briarane diterpenes (Qi et al. 2006, 2009) revealed from the gorgonian *Junceella juncea* located in South China Sea, were extremely antifouling concerning cyprids of *Balanus amphitrite*; (**8**) exhibited ED_{50} value 0.004 $\mu\text{g/mL}$ and (**9**) exhibited ED_{50} values 0.005 $\mu\text{g/mL}$; however, (**10**) was active merely (ED_{50} valued 22.6 $\mu\text{g/mL}$).
- (h) Three meroditerpenes (**11**), (**12**), (**13**) derived from *Halidrys siliquosa* which is a brown alga located in Mediterranean exhibited antifouling properties against cyprids of *Balanus amphitrite* and exhibited EC_{50} measure of 1, 5, and 4 $\mu\text{g/mL}$, correspondingly (Culioli et al. 2008). Noticeably, the initial two compositions (**11**, **12**) were very lethal to naupliar larvae (LD_{50} valued 5 and 10 $\mu\text{g/mL}$, correspondingly); however, (**13**) exhibited its non-toxic nature even at 100 $\mu\text{g/mL}$.
- (i) A total of fourteen terpenoids derived from sponges located in the Mediterranean were assessed for antifouling action and noxiousness concerning larvae of barnacle. Out of which the capable compounds were hydroquinone C acetate (**14**) and dihydrofurospingin II (**15**) whose ED_{50} values were less than 1 and 2.5 $\mu\text{g/mL}$, correspondingly and LD_{50} value was greater than 100 $\mu\text{g/mL}$ (Hellio et al. 2005).
- (j) Steroid (**17**) a 5 α -hydroxylated clone of (**16**) which was derived from gorgonian *Subergorgia suberosa* located in South China Sea exhibited strong antifouling activity against *Bugula neritina* and *Balanus amphitrite* (Qi et al. 2010).
- (k) Firsthand A-nor steroids, like (**18**), derived from *Acanthella cavernosa* a oceanic sponge, exhibited reasonable antifouling against barnacle *Balanus albicostatus* (EC_{50} values ranging 8 to 32 $\mu\text{g/mL}$) (Qiu et al. 2008).

Fatty acids and associated combinations

- (a) Aureobasidin (**19**), an uncommon bis-ester derived from the genus *Aureobasidium* of a marine fungus, exhibited capabilities of hindering the larval establishment of barnacle *Balanus amphitrite*, but research data is unavailable (Abdel-Lateff et al. 2009).
- (b) (3*R*,5*S*)-3,5-Dihydroxydecanoic acid (**20**) likewise reported analogous activity of inhibiting larval estab-

lishment of *Balanus amphitrite* (Abdel-Lateff et al. 2009).

Amino acids and related compounds

Bastadins is a class of bromotyrosine derivatives of compounds found in marine sponges and were antifouling against *Balanus amphitrite*.

- (a) Among the eighteen number of bastadins and related compounds that were being tested, the ultimate active compound was Bastadin-9 (**21**) with a least possible inhibitory concentration measuring about 1.0 μM , although these compound were nonhazardous concerning larvae of barnacle at 10 μM . (Ortlepp et al. 2007).
- (b) A truncated bastadin derivative known as Hemibastadin-1 (**22**) was highly active (minimum inhibitory concentration of 10 μM), and also was fatal to cyprids at valued of 1 μM .
- (c) Debromohemibastadin-1 (**23**) demonstrated significantly lesser antifouling activity (minimum inhibitory concentration of 100 μM).
- (d) Artificial dibromohemibastadin-1 (**24**) and natural aplysamine-2 (**25**) exhibited decent barnacle resistant nature (minimum inhibitory concentration of 10 μM); however, both were nonhazardous.
- (e) Brominated diketopiperazines (DKPs) of two in numbers derived from the aquatic sponges *Geodia baretii* of Swedish origin namely, Baretin (**26**) and 8,9-dihydrobaretin (**27**) had shown strong antifouling action concerning the barnacle species *Balanus improvises*. They demonstrated EC_{50} measure of 0.9 and 7.8 μM , correspondingly. Notably, the aftermath of the compounds was alterable and nonhazardous (Johnson et al. 2004; Sjögren et al. 2004b).
- (f) The studies of structure–activity relationship using artificial derivatives of baretin facilitated revelation of benzo[g]dipodazine (**28**), which exhibited reversible activity and EC_{50} valued of 0.034 μM (Sjögren et al. 2006). A 2-month real-life ground test with a polymer comprising 0.1% dry weight of baretin revealed a decline in the establishment of barnacle by 89% and establishment of mussel by 81% (Sjögren et al. 2004a).
- (g) An innovative protease made by the bacterium *Pseudalteromonas issackenkonii* derived oceanic sediments subdued establishment of larvae of the bryozoan *Bugula neritina* with EC_{50} valued 0.5 ng/mL (Dobretsov et al. 2007). An in situ experiment involving inclusion of this enzyme in water soluble paints displayed a noteworthy inhibition of establishment of both barnacles and bryozoans (Xiong et al. 2007).

Benzenoids

- (a) Polybrominated diphenyl ethers derived the genus *Dysidea* of aquatic sponges and the artificial analogs of the same were assessed for antifouling action concerning larvae of barnacle. The compound (**29**) was revealed to be the most active with EC_{50} valued 1.26 μM (Ortlepp et al. 2008).
- (b) The metabolic product of an aquatic fungus *Ampelmyces* is 3-Chloro-2,5-dihydroxybenzyl alcohol (**30**) naturally obtained from biofilms was assessed. It repressed the establishment of larvae of barnacle *Balanus amphitrite* (EC_{50} value ranging between 3.19 to 3.81 $\mu\text{g/mL}$ and LC_{50} measure of 266.7 $\mu\text{g/mL}$) (Kwong et al. 2006).
- (c) Luteolin-4^l-glucuronide (**31**) is a flavonoid glycoside derived the seagrass *Enhalus acoroides*. It repressed the establishment of larvae of barnacle *Balanus amphitrite*. The EC_{50} valued 0.52 $\mu\text{g/mL}$, and demonstrated a very minute toxic nature (Qi et al. 2008a).

Alkaloids

A combination of araguspongines containing araguspongine C (**61**), a bis-1-oxaquinolizidine alkaloid-derived *Haliclona exigua*, an aquatic sponge of Indian origin, repressed adhesion of cypris of *Balanus amphitrite* (EC_{50} valued 6.6 mg/mL); however, exhibited toxicity (LD_{50} 18 mg/mL) (Limna Mol et al. 2009).

Unspecific categories

Apart from compositions of specific categorization like amino acid-associated compounds, fatty acid-linked compounds, and benzenoids, several unspecified compositions extracted from marine and terrestrial organisms demonstrated barnacle-resistant nature which are the following:

- a. Alkylated butenolides (**33,34,35**) derived an actinomycete *Streptomyces* originated in deep oceanic residues. Although it demonstrated a weak toxic nature (the value of LD_{50} greater than 100 $\mu\text{g/mL}$) but displayed effective antifouling action concerning larvae of *Balanus amphitrite* and the EC_{50} value ranged between 8 and 15 $\mu\text{g/mL}$ (Xu et al. 2010).
- b. The study of alkyl butenolide's structure and activity association reveals that the ring of 2-furanone was vital for the antifouling action, and considerable contribution from the side chain's lipophilic nature was noted. This benefited the synthesis of a standard alkyl butenolide (**36**), which displayed capable antifouling activity (EC_{50} valued 0.5 $\mu\text{g/mL}$ and LD_{50} greater than 50 $\mu\text{g/mL}$ for *Balanus amphitrite*) (Xu et al. 2010).

- c. Maculalactone A (**37**) is a tribenzyl butenolide derived the aquatic cyanobacterium *Kyrtuthrix maculans*, was found to be fatal to naupliar larvae of numerous barnacles, with LD₅₀ values ranging from 1.1 to 5.2 µg/mL (Brown et al. 2004).
- d. Floridoside (**38**), derived the red alga *Grateloupia turuturu*, exhibited effective anti-barnacle action (EC₅₀ value less than 1 µg/mL); however, the same behaved harmlessly to naupli at a concentration of 10 µg/mL. Notably, isethionic acid (**39**) exhibited contrasting characteristic, that is minor dynamic nature, but more lethality (Hellio et al. 2004).
- e. Manoalide (**40**) derived the aquatic sponge *Luffariella variabilis* demonstrated suppressive effects on larvae of barnacle *Balanus amphitrite* at small proportions (EC₅₀ 0.24–0.80 µg/mL) (Tsukamoto et al. 1997).
- f. For assessment of antifouling nature applicable to larvae of *Balanus amphitrite*, an aggregate of seventeen flavonoids and isoflavonoids were analyzed. This resulted in the unearthing of a capable and safe antifoulant candidate called genistein (**41**), exhibiting EC₅₀ valued 3.0 µg/mL and therapeutic rate more than 16.6 (Zhou et al. 2009). Genistein also unveiled capable antifouling nature in an initial ground experiment. Also, insecticides derived from plants like pyrethroids exhibited capable antifouling activity.
- g. Cypermethrin (**42**) repressed the larval establishment of *Balanus albicostatus*. It exhibited EC₅₀ measure of 0.11 µg/mL and LC₅₀ measure was greater than 250 µg/mL (Feng et al. 2009a). Interestingly, it demonstrated its activeness as tributyltin in a field experiment.
- h. 3,7-Dimethyloctyl 2-methyl-5-pyridyl ether (**43**) repressed establishment of larvae of cypris of *Balanus balanoides*, and an 8-month-long in situ experiment ensured that nil barnacle attachment on the exterior of the substrate. It exhibited LC₅₀ measure 6 µg/mL against *Balanus balanoides* (Skattebøl et al. 2006).
- i. Polyphenolic compounds such as tannins are generally widely dispersed in plants and exhibit a broad range of organic activities. Notably, three compositions of tannins revealed that they are harmful to naupliar larvae of barnacle *Balanus amphitrite*, and a real-life experiment with soluble matrix paints prohibited establishment of barnacles (Stupak et al. 2003). During real-life experiment, performance of quebracho tannin as well as aluminum tannate was noticeable because they entirely restrained the motion of larval attachment of barnacles and polychaete at little proportions, and lessened the fouling effectively within a months' time (Pérez et al. 2007).
- j. The ultimate promising antifouling enzymes are proteases, like the serine protease, Alcalase1, which had demonstrated degradation of cement of barnacle cyprid and averted larval connection (Aldred et al. 2008). Subtilisin A is the vigorous ingredient of Alcalase which arrested the matrix of a polymer and exhibited reduction of the establishment and bonding power of algal cells (Tasso et al. 2009).
- k. Zosteric acid or p-(sulfoxy) cinnamic acid (**44**) being a natural produce from *Zostera marina* which is a seagrass (Todd et al. 1993) exhibited an anti-biofouling activity (prominently against bacteria and barnacles) (Callow and Callow 1998; Barrios et al. 2005; Villa et al. 2010, 2011, 2012). It also exhibited very little toxicity (EC₅₀ value ranging between 170 and 450 ppm) (Xu et al. 2005; Villa et al. 2011; Cattò et al. 2015). The approach of antifouling action is yet to be detailed publicly; however, prediction was that zosteric acid may behave as an ecological signal, which resulted in overall pressure on the cells and supported the expression of numerous proteins obligatory to discharge from hostile circumstances (Villa et al. 2012; Chen and Qian 2017).
- l. 2'-Methoxykurarinone (**48**) derived the plant *Sophora flavescens* of medicinal value, repressed settling of cyprids of *Balanus albicostatus*. The EC₅₀ measure was 2.0 µg/mL and its injuriousness was on the lower side (LC₅₀ > 25 µg/mL) (Feng et al. 2009b).
- m. Matrine (**49**), also derived the plant *Sophora flavescens* of medicinal value, repressed settling of cyprids of *Balanus albicostatus*. The EC₅₀ measure was 7.1 µg/mL, while its harmfulness was lesser (LC₅₀ > 250 µg/mL) (Feng et al. 2009b).

Techniques to boost production of marine natural antifoulants

The discovery of a principal formulation through the laboratory screening is very exhilarating but the requirement of large quantities of natural antifoulants for field assays and formulations for paints, their complications of massive manufacturing became a serious constriction (Hellio et al. 2009). Numerous techniques for a sustainable production of natural antifoulants are outlined in Fig. 11. However, only for pharmaceutical applications, these natural products have been only scaled up and not for antifouling preparations in spite of the availability of all these expertise (Marris 2006). Even if no natural products had made it yet to the industrial application of fouling resistant coatings (Pawar et al. 2020), an interesting product of polymer kind named Biojelly® (Hayase et al. 2003) was registered for patent few years back. It was formed on a membrane composed of cellulose acetate which was submerged in saltwater and harbor specific bacteria, which impeded the adhesion of aquatic creatures such as algae and barnacles (Hayase et al. 2003). During the release in the environment most of the secondary metabolites are

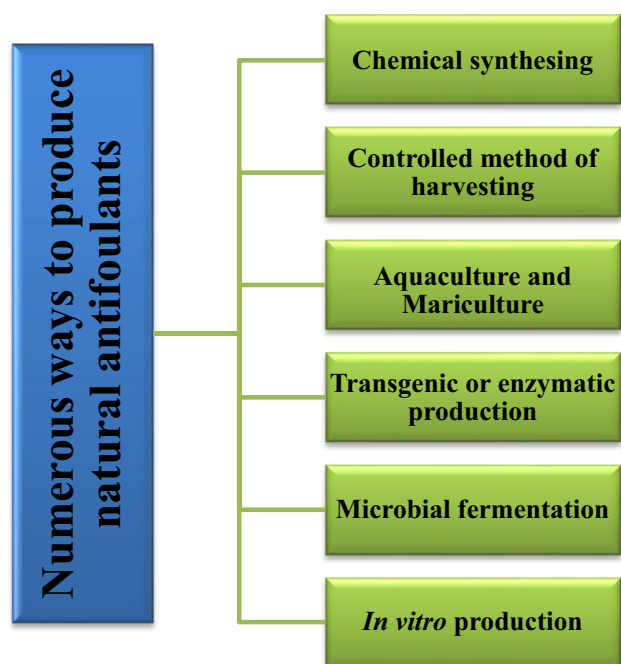


Fig. 11 Methods to produce natural antifoulants

rapidly broken down (Price et al. 1992) and hence their integration in a formulation of paint is very challenging. For enhancing the durability of the coatings, regulation of the rate of release has to be monitored and microencapsulation of the bioactive natural products is the most suited method (Price et al. 1992).

Synthetic compounds

1-Alkyl-3-methylimidazolium ionic liquids

Ionic liquids are liquid salts having melting point below 100 °C and constitute equimolar concentrations of cations and anions (Rogers and Seddon 2003). Among the different classes of ionic liquids, alkyliimidazolium ionic liquids were extensively studied with a large number of reports on synthesis, physicochemical properties, and applications (Olivier-Bourbigou et al. 2010). In addition to diverse industrial applications (Vekariya 2017), ionic liquids are being explored for their utilization in the biomedical and pharmaceutical industries (Egorova and Ananikov 2018). Antimicrobial and antibiofilm characteristics of alkyliimidazolium ionic liquids had been widely reported against a large number of bacteria (pathogenic and environmental), fungi, and drug-resistant microbes (Venkata Nanchaiah et al. 2012; Bergamo et al. 2015; Doria et al. 2018). Alkyliimidazolium ionic liquids revealed that they exhibited suppression capabilities on the growth and biofilm formation by phototrophic

organisms (Reddy et al. 2017). Alkyliimidazolium ionic liquids impede development and biofilm establishment by permeabilizing the cell membranes in bacteria and microalgae (Venkata Nanchaiah et al. 2012; Reddy et al. 2017; Egorova and Ananikov 2018).

The anti-barnacle nature of 1-alkyl-3-methylimidazolium ionic liquids ($[C_nMIM]^+[X]^-$, $n=4, 12$, and 16) was assessed with the help of barnacle larvae. Survival and establishment of cypris larvae were determined with the help of different concentrations of three different ionic liquids. The experimental results demonstrated that the tested 1-alkyl-3-methylimidazolium ionic liquids are very useful to prevent the establishment of barnacle larvae. The establishment of cypris larvae was strongly inhibited in presence of nanomolar concentrations of tested compounds suggesting their potential use in the preparation of antifouling formulations/paints.

The tested compounds are:

- (a) $[C_4MIM][Cl]$: 1-butyl-3-methylimidazolium chloride
- (b) $[C_{12}MIM][I]$: 1-dodecyl-3-methylimidazolium iodide
- (c) $[C_{16}MIM][Cl]$: 1-hexadecyl-3-methylimidazolium chloride

The tested alkyliimidazolium ionic liquids with *-butyl*, *-dodecyl*, and *-hexadecyl* side chains exhibited antifouling potential. These compounds were found to cause mortality in cyprid larvae at micromolar concentrations and also prevented establishment at nanomolar concentrations. Alkyliimidazolium ionic liquid compounds also affected the survivorship and inhibited the metamorphosis of different naupliar larval stages of barnacle. The EC_{50} concentrations obtained for all the three tested ionic liquids ($[C_4MIM][Cl]$, $[C_{12}MIM][I]$ and $[C_{16}MIM][Cl]$) were 14, 0.147, and 0.535 $\mu g/L$ respectively) which were noticeably much lesser than the reference value (25 mg/L) to qualify for a potential antifouling agent (Almeida and Vasconcelos 2015). This indicated the effectiveness of the tested compound; thereby, requirement of minimal concentrations of the tested compounds is sufficient to impede the establishment of cyprid larvae. Besides, the therapeutic ratios ($T_R = LC_{50}/EC_{50}$) of the experimented compounds ranged between 28 to 900 which were much greater than the reference value (> 15) as per the benchmark requirement set by the US Navy for an effective antifouling agent (Almeida et al. 2017), thereby qualifying these as ecologically promising and safe antifoulants. Out of the three experimented ionic liquids, $[C_{12}MIM][I]$ displayed best anti-biofouling activity with a cypris mortality (LC_{50}) valued of 0.35 μM and establishment inhibition (EC_{50}) valued of 0.39 nM in comparison with the other two compounds. $[C_{16}MIM][Cl]$ displayed medium activity with (LC_{50}) valued of 0.63 μM and (EC_{50}) value

of 1.56 nM in comparison with the other two compounds. Noticeably [C₄MIM][Cl] displayed the least activity on survival and establishment with (LC₅₀) valued of 2.28 µM and (EC₅₀) valued of 80.2 nM. Noticeably, there was no contribution by the anions (Cl[−] or I[−]) to the biological activity of these liquids.

The outcome of the experimented ionic liquids on killing and metamorphosis of nauplius larvae followed the following decreasing order: [C₁₂MIM][I] > [C₁₆MIM][Cl] > [C₄MIM][Cl]. The killing, transformation inhibition, and establishment inhibition activities of alkylimidazolium ionic liquids increased with a surge in length of alkyl chain from -butyl to -dodecyl group. However, [C₁₂MIM][I] with -dodecyl alkyl chain was found to be more potent than [C₁₆MIM][Cl] with -hexadecyl alkyl chain in exhibiting antifouling activity.

Poly(L-lactic acid)

Literature surveys established that certain ecofriendly polymers (Tokiwa and Calabria 2007) such as poly(3-hydroxyalkanoate)s (Hany et al. 2004), poly(ε-caprolactone) copolymers (Faÿ et al. 2006), poly(ester-anhydride) (Faÿ et al. 2007a), poly(lactic acid) (Faÿ et al. 2008), graft copolymers containing poly(lactic acid) side chains (Langlois et al. 2002), and poly(ε-caprolactone-co-lactide) (Faÿ et al. 2007b) had been used as ecologically harmless surface-fragmenting or self-polishing matrices to control the discharge of the antifoulants. The rate of discharge of biocide was regulated by the volume of oligo (D, L-lactic acid) units present in the graft copolymers (Langlois et al. 2002). On the contrary, the lesser the molar mass of poly(ε-caprolactone-co-lactide), the rapid would be the release rate of biocide from antifouling paints, resulting in a slowdown of the antifouling activity (Faÿ et al. 2007b).

The property of releasing of lactic acid and the ratios of affixations as well as durability of larvae of cypris of *Balanus amphitrite* were examined by preparing testers of poly(L-lactic acid) sheet of varying molar masses. Here, testers with applied poly(L-lactic acid) on the exterior were immersed in seawater for investigation. It was observed that the rate of release of lactic acid from the testers was reciprocally proportionate to the molar mass of poly(L-lactic acid) despite the prevailing buffering effect of seawater. The reduction in the affixation ratio of larvae of cypris with the rise in the rate of discharge of the lactic acid poly(L-lactic acid) demonstrated the anti-barnacle attachment property of poly(L-lactic acid). The adding of lactic acid directly into the laboratory culture test also exhibited an analogous repression consequence on the attachment of cypris larvae (Ishimaru et al. 2012).

Thymol, eugenol, and guaiacol as bioactive additives along with natural phenolic compounds and low copper content

The development of leading biocide in massive proportion for commercial manufacture of antifouling paints was cuprous oxide and presently this is comprehensively used in tributyltin-free antifouling products (Cima and Ballarin 2012). However, both the presence of high concentrations of copper in aqueous medium (like complex forms or free cations of cuprous or cupric), sediments, and biotic matters near to harbors and the growing pressure of regulation on copper oriented paints had led to widened exploration of substitute for ecologically safe antifoulants (Champ 2000; Srinivasan and Swain 2007). Despite its greater efficacy, algal fouling is more resistant to cuprous oxide than animal fouling. This has led researchers to probe for additional biocides, called “booster biocides,” which when combined with cuprous oxide improves its efficiency. Therefore, the antifouling paints which are present in commercial scale have cuprous oxide within a polymeric framework solely or, in amalgamation with other organic promoter compositions, to escalate the effectiveness and target range of antifouling coatings. Notably, lethal effects on ascidians embryos can be caused by booster biocides (Gallo and Tosti 2015). A number of researches have assessed the deadliness of booster biocides on nonspecific organisms and have established that many of these were growth suppressers for freshwater and aquatic organisms (Okamura et al. 2003) manipulating vital species, such as seagrasses (Chesworth et al. 2004) and corals (Owen et al. 2002). Therefore, the ban on usage of these booster biocides was anticipated due to the rising apprehension about the fate of the environment and the potent risks (Guardiola et al. 2012).

Thymol (45), eugenol (46), and guaiacol (47) are existent in the necessary oils as clove oil, thyme oil, and cinnamon which are derived different herbs and spices, and their bacterial-resistant and fungal-resistant nature are well known.

Thymol is one of the chief constituent of thyme oil, which is widely known for its medicinal nature, along with microbe resistant, fungal resistant, insecticidal, and antioxidant effects (Cimanga et al. 2002; López-Malo et al. 2002; Tepe et al. 2005; Braga et al. 2007; Dalleau et al. 2008).

Eugenol is a phenylpropanoid which is an allyl chain-substituted guaiacol. It exhibits weak acidic nature, slight solubility in aqueous medium, and soluble in organic medium. It also exhibits a transparent to pale yellowish colored fluid with a distinctive and pleasurable aroma of cloves, a sharp spicy flavor, fungal-resistant properties (López-Malo et al. 2002; Kamatou et al. 2012), and is a quorum-sensing inhibitor (Zhou et al. 2013). Eugenol for business purposes is solely derived sustainably

from natural sources. It is also manufactured artificially by allylation method between guaiacol and allyl chloride (Kamatou et al. 2012). Generally, the massive manufacturing of thymol and guaiacol is by biological synthesis (Amandi et al. 2005; Fu et al. 2005) and biotechnology processing (Witthuhn et al. 2012).

Guaiacol (o-methoxyphenol; methylcatechol) is a natural composition derived from the resin of *Guaiacum* trees (Parimal et al. 2011) and essential oil of many terrestrial plants like tobacco (Rodgman and Perfetti 2008).

The existence of a hydroxy (-OH) group, affixed to a benzene ring or other complex aromatic ring configurations, like resorcinol or pyrogallol or catechol denotes phenolic compounds. The best microbial-resistant vital constituents are the naturally occurring phenolic substances which are present in plant as essential oils. These demonstrated a shallow lethal effect on homeothermic animals in spite of greater antifungal, antibacterial, and insecticidal efficacy of these compounds (Zabka and Pavela 2013). It was revealed that phenols exhibited effective antioxidant nature (Ruberto and Baratta 2000). Notably, the antioxidant character of the plant oils are mainly due to oxygenated monoterpenes like thymol. Thymol, eugenol, and guaiacol were chosen as capable contenders for impeding marine biofouling because of the presence of these phenolic structure.

The healthy larvae (nauplii II) of *Balanus amphitrite* sank to the bottom of the vessel when these came in contact to either thymol, eugenol, and guaiacol, indicating that *Balanus amphitrite* larvae were vulnerable to these compounds. The establishment as well as the toxicity inhibition tests revealed that the effectiveness of these compounds widely varied, despite being structurally similar compounds. Thymol and eugenol in low concentrations affected naupliar activity. Nauplii lost phototactic reaction and motile activities which were detected for thymol solutions on concentrations greater than 3 μM and for eugenol solutions on concentrations greater than 0.048 μM . The value of LC_{50} for thymol is 4.41 μM and for eugenol is 0.059 μM . Despite the presence of potent inhibition capability of guaiacol concerning the larvae, the effective proportion required for attaining dormant state was 0.364 μM and LC_{50} value 0.53 μM ; in other words, thrice greater than thymol and eugenol (Pérez et al. 2015).

Polydimethylsiloxane-based fouling release coatings with the integration of non-reactive phenylmethylsilicone oil

In the year 1977, first report of polydimethylsiloxane-based fouling release coatings with the integration of non-reactive phenylmethylsilicone oil (PSO) was available. The main objective for integration of non-reactive silicone oil was to

mimic the exterior secretion on the fish skin likewise release of silicone oil onto the coating surface leading to augmentation of the antifouling performance. Few progresses related to the above are globally available (Truby et al. 2000; Stein et al. 2003; Galhenage et al. 2016; Shivapooja et al. 2016). The integration of silicone oil improves the greasing nature of the surface of the coating thereby degrading the affixation power of the foulants to the exterior of the substrate. The integration of phenylmethylsilicone oil along with polydimethylsiloxane based fouling release coatings displayed outstanding performance concerning barnacle adhesion resistant (Truby et al. 2000), due to the reduction of strain of substratum required for barnacle detachment (Shivapooja et al. 2016). The lesser toxic nature of the dominating silicone oil is not a menace straightforwardly to the living species and also there is no statistical evidence to support threat of marine environment pollution due to the leached silicone oil existence (Truby et al. 2000). The introduction of hydroxylated multi-walled carbon nanotube ferric oxide [(MWCNTs-OH) / Fe_2O_3] has also influenced the performance of coating. With an escalation in the volume proportion of hydroxylated multi-walled carbon nanotube in the mixed particle, it was observed that there were reduction of the smoothness of coating surface and enhancement of the mechanical properties. A transition of the percolating behavior of phenylmethylsilicone oil that is from the diffusion in molecular network to the channelized diffusion was also revealed. Initially, the efficiency of leaching of phenylmethylsilicone oil weakened but was improved due to the alteration of percolating behavior. The outcome of the adhesion of the biofilm analysis highlighted the performance of antifouling nature of the coating was primarily dependent on the leaching of phenylmethylsilicone oil. Hence, it is imperative to regulate the percolating of phenylmethylsilicone oil. The antifouling panel test highlighted that if the volume proportion of hydroxylated multi-walled carbon nanotube in the mixed particle was greater than 50% then achievement of excellent antifouling performance of the coating could be ensured (Ba et al. 2019).

Polydimethylsiloxane-centered films comprising surface-active amphiphilic block copolymers

Several strategies in production of innovative coatings of fouling-resistant and fouling release nature have been analyzed that combine diverse polymer compositions, along with self-organized copolymers with properties like polymer nanocomposites (Korde et al. 2020; Jaya Prakash and Kandasubramanian 2021; Suraj Belgaonkar and Kandasubramanian 2021) polymers of zwitterionic nature, phase-segregated polysiloxane-urethanes, and side chains of mesogenic nature (Oliva et al. 2017). The present promising strategy to combat marine biofouling is utilization

of amphiphilic polymer films, comprising of both water-absorbing and water-repelling characteristics on the same surface. Various chemical methods, like silica-reinforced silicones with poly(ethylene oxide)-polysiloxanes curing blends, alteration of polymer scaffolds by glycine polypeptides, combinations of polyethylene glycol (PEG) along with fluorinated big-sized monomers being photo-cross-linked by ultraviolet rays, cross-linkages between polyethylene glycol with hyperbranched fluoropolymers introduce amphiphilic characteristics in materials. M. Oliva et al. probed the introduction of block copolymers of amphiphilic nature in matrix polymer films as surface-active agents and to avert fouling, the role of macromolecular engineered modified coating surfaces. The twin fold chemical nature enables the reorganization of the surface and modification of its composition in accordance with external circumstances, hence averting the establishment or stimulating the elimination of foulants. A firsthand set of surface-active block copolymers was produced which combined a first block of polysiloxane and a second block of poly (4-triethyleneglycol monomethyl ether)-2,3,5,6-tetrafluorostyrene). For establishing their competence to diminish the establishment or to enable the removal of two specific barnacle species, this was dispersed in a polydimethylsiloxane matrix. The technique of atom transfer radical polymerization (ATRP) of the amphiphilic monomer 4-(triethyleneglycol monomethyl ether)-2,3,5,6-tetrafluorostyrene (EFS) starting from one same bromoterminated polydimethylsiloxane macroinitiator (Si-Br, number average degree of polymerization 11) was deployed for production of the two block copolymers Si-EFS14 (degree of polymerization 14) and Si-EFS71 (degree of polymerization 71). The proportion of settling of cyprids of *Balanus improvisus* on investigating experimental surfaces were more in comparison to that of *Balanus amphitrite*. But the elimination of the younger *Balanus improvisus* from the coating of the copolymer Si-EFS14 was easier in comparison to that of the copolymer Si-EFS71. The enrichment of the nearby surface of Si-EFS71 copolymer with oxyethylenic chains after immersion in water was also observed through x-ray photoelectron spectroscopy (Oliva et al. 2017).

Fouling release coatings

The progress in non-hazardous lesser surface energy fouling release coatings (FRCs) is being lately comprehensively investigated. The foul release coatings exhibit as a capable non-lethal substitute to coatings comprising of biocide or lethal compounds. The minimization of the inter-molecular forces during contact between the fouler and the substrate ensuring their effortless release during low shear stresses is the key design methodology for novel foul release coatings. The low modulus and the low surface energy characteristic of PDMS- or silicone-based FRC permit the adhered foulers

to get freed from the surface effortlessly. Even though the colonization of foulers can occur on this but these can be detached with ease either by scrubbing or by self-cleaning due to hydrodynamic forces (Rath et al. 2018). The hydrodynamic self-cleaning technique is when the speed of the vessel generates adequate shear for detaching the foulants. Hence, when a vessel is operational in seawater, the fouling release characteristic of these coatings are found to be commendable (Rath et al. 2010).

A diversity of silicone coatings of fouling release nature for protection of exteriors have been evaluated by numerous investigators. The investigations revealed that cautious design of several molecular architectures and network formation using PDMS has exhibited considerable detachment of foulants but these coatings are delicate and vulnerable to mechanical harm and exhibited deficiency of adhering to various other substrates. In order to evade these problems, development of PDMS-modified polyurethane (PU) and epoxy resins was undertaken and assessed to augment the adhering capability and resilience of FRCs. The mechanism of self-stratification is the backbone of both the urethane and epoxy modifications of silicones where during the film formation, the constituent having lesser surface energy enriches the air interface before it becomes a constituent part of the polymer network (Rath et al. 2018). The amalgamation of properties like higher thermal stability, lesser surface tension, and lower glass transition temperature of silicones has made this commercial polymer very useful despite its comparatively meager mechanical strength. The high mobility of the chains permits the expansion of the flaws or microcracks which result in the meager strength. Abundant artificial techniques have been endeavored for augmentation of the mechanical characteristics of the silicone materials while retentive of the in-built properties of silicone moiety. The approaches implemented are usually combination of siloxanes with hydrocarbon-based polymers, specifically resins of epoxies (Rath et al. 2009).

S.K. Rath et al. (2010) investigated the FRC property of PDMS. The specimens were layered with silicone-incorporated epoxy resins and were dipped at a water basin off the Mumbai harbor to assess the effectiveness of the bonding of fouler. The preliminary settlement of the fouling organisms was visualized after a fortnight of exposure in the sea and primary observations were recorded. Afterwards, the same specimens were retrieved after 3 months for macrofouling visualizations and recording. The control specimen along with the silicone incorporated epoxy-coated specimen was entirely wrapped by slime and establishment of barnacles, polychaetes, oysters, and ascidines. Relatively, the silicone-incorporated epoxy-layered specimen displayed considerable enhanced fouler detachment nature than the uncoated specimen during jetting with water from a hose pipe. The 15% silicone-incorporated epoxy blend displayed 25 to 35%

debonding of the foulants, in contrast to the 30% silicone-incorporated specimen where a 50 to 60% debonding of the foulants was detected. The silicone-incorporated specimen exhibited substantial enhancement in detachment of foulants in comparison to the control specimen. Yet hard shell foulers were still prevalent even after cleaning with water. The predominance of barnacle establishment in contrast to hydroids and oysters on the coated panels were also highlighted.

Safe environmental techniques for mitigation of biofouling

A balanced overview of firsthand biocide-less and fouling-release coatings engaging ecologically concerned surface-active interactions and patterns of engineered micro-topographic nature was seen in response to constraints on usage of biocides in antifouling coatings. This was aimed to either discourage establishment of larvae of cyprid or to diminish the adhesive nature of adhered foulants so that the surfaces already fouled can be scrubbed with ease. Though these approaches have been publicized to repress adhesion of foulants, but hardly these achieve complete fouling-free condition. Numerous chemical methods for elimination of the attached hard foulers had been developed and employed to boost the efficacy of fouling resistant coatings. The techniques of chemical in nature like scrubbing with solution of acidic nature have been deployed for removal of barnacles in spite of the knowledge of the sturdy potent caustic nature of acidic blends are perilous for usage and lead in noxious releases.

Few ecofriendly techniques for cleaning of hard foulants are enumerated below:

(a) Impact of cavitation generated bubbles

S. Guo et al. (2013) highlighted the impact of cavitation bubbles on getting rid of juvenile barnacles. The liquid jet as a consequence of the collapse of bubbles was perceived to be oriented toward the hard foulants at diverse impact intensities. The main parameters that played a crucial role are the distance between point of formation of the bubble and the exterior of the barnacle, and the maximum radius that the bubble can attain. The photography using a high-speed camera recorded the interaction between bubbles and barnacles to understand the phenomenon. The pre and post state of the barnacles also highlighted the impact of cavitation. The ultrasonic cavitation led liquid jet injured the fragile shells of barnacles with ease. Various exposure timings of 30 s, 60 s, and 150 s caused partial damage of shells, full damage of shells, and full removal barnacles correspondingly. It is predicted that biological damage may be induced by that micro-streaming and shock waves and ultrasonic cavitation may

produce forces by liquid jet. It is understood that collapsing of bubbles produces impulsive pressure which causes cavitation damage. The bubbles induced by ultrasonic cavitation collapses nearby a boundary, resulting microjets to penetrate through the cavity and imping on the boundary with great amount of pressure. The disengagement of bacteria and biofilms occurs by impingement of liquid jet which is produced by the cavitation. The impairment to the surface was caused by the shear force generated by the radial flow which was prominent along with the influence of liquid jet. The thorough elimination of the adhered barnacles was performed by bubbles of moderately smaller dimension, whereas elimination of the older barnacles required greater force. The impact of bubble cavitation was inadequate to cause delamination because of existence of cement of barnacles; however, with greater force in the form of a bigger bubble scrubbing of the shell of the older barnacle was achieved. This research concludes that influence of ultrasound cavitation is effective only during the preliminary stages of fouling and might be inefficient if applied to older barnacles.

(b) Impact of ultrasound

S. Guo et al. (2011) probed the suppressing consequence of ultrasound on barnacle cyprids. The ultrasound-induced effect can lead to biological alterations due to ultrasonic pressure fluctuations. The influences like radiation torque and force, micro-streaming induced by ultrasound wave can lead to the distortion and upsetting of biotic materials, decomposition of biotic structures, and augmenting of enzymatic reactions. Of the three frequencies, the utmost capable frequency is 23 kHz, whereas analogous but feebler response was obtained from 63 and 102 kHz. The effect of reduction of establishment was obtained with greater ultrasonic irradiation and a lower range of frequency (around 20 kHz) had more lethal influence on naupli of barnacles. The decrease in establishment revealed that the ultrasonic cavitation caused bodily damages on the cyprids. The presence of smaller bubbles in the liquids would oscillate or implode due to cavitation leading to greater shear stress and liquid jets that could physically injury proximate creatures. The existent biotic species are harmed due to greater force of shear stress which was probably caused by the bubble led micro-streaming in the nearby liquid. For suppression of bacterial growth, removal of algae, and destruction of biofilm, ultrasound cavitation induced shear stress, shock wave, and liquid micro-jets are responsible. Even for treatment of ballast water by destruction of larvae of barnacles, significant cavitation is caused by ultrasound of higher intensity. The acoustic pressure value should be greater than 5 kPa for ensuring the continuation of the suppression effect. For repression of establishment of cyprids, the technique of ultrasound is a promising one and it holds numerous benefits

in comparison to biocide-centered processes. The prime benefit in application of ultrasound over biocide based is that the later may leach out continuously; however, application of ultrasound can be regulated and monitored.

(iii) Impact of ultraviolet light

K. Hunsucker et al. (2019) piloted a two phased assessment to investigate the efficacy of using ultraviolet light (200–280 nm), along with ship hull coatings. This combination could be a probable approach to avert biological fouling by boosting the coating's performance. The first stage of research estimated the frequency of ultraviolet light required for additive free coating. The second stage of research highlighted several elements like various types of coatings (epoxy-oriented coatings, copper-oriented coatings, fouling release type of coatings), frequency of ultraviolet light (zero exposure, uninterrupted exposure, exposure for 1 min per 6 h, exposure for 1 min per day), and gap from the lamp (25 mm and 50 mm). The successful application of ultraviolet light for the deterrence of biofouling in salt-water arrangements, on appliances and sensors, has been documented and presently its application on vessel exterior surfaces is being investigated. For inhibition of bacterial biofilms, and in-water treatment facilities UV light technique is already applied. The maximum germicidal nature exhibited by UV light is in the array of 100–280 nm of UV wavelength, because it disrupts within the microorganisms the DNA–RNA polymers' chemical bonds. It was witnessed that both for copper and silicone-centered fouling control coatings, the exposure of minimum 1 min per day was adequate to avert fouling; however, for epoxy centered, the same was unsatisfactory. There was presence of deterioration in copper-centered coatings which were exposed to incessant ultraviolet light. An augmented proportion of hydrolysis within the coating might be due to probable exposure of ultraviolet light. The possibility of molecular level interactions could not be ruled out; however, there was no evident impairment on the silicone or epoxy coatings. Additional extensive investigation would probably expose the influences of the ultraviolet on these coatings and the prospective influence on the fouling leaching characteristics of silicone. During executing ultraviolet systems in the aquatic environment, the influence of turbidity is one of the chief considerations because an amplified proportion of turbidity can decline the transmission of ultraviolet light and lessen its efficacy on the targeted surface. The variation of turbidity is affected by both environmental circumstances as well as topographical position. The modification of the dose–response diagram could also guard microorganisms from the ultraviolet light. It has been observed that lesser diffusion through seawater (60%) leads to lesser threshold strength for averting biofouling establishment. The expanse amid the ultraviolet light source and the targeted surface would also influence the

surface area of exposure. The lenience nature in the foulant community also highlighted disparities concerning the composition of fouling.

(iv) Impact of pulsed laser irradiations

K Nandakumar et al. (2002) examined the efficacy of pulsed laser irradiations from an neodymium doped yttrium aluminum garnet (Nd:YAG) laser of second harmonic that created a light irradiation of 532 nm to impair the larvae of *Balanus amphitrite* in a simulated surroundings. The established average fluence was 0.1 Jcm^{-2} . A total of larval stages of three numbers were exposed to irradiations of laser for varying time period. The recorded pulse width of 5 ns, repetition rate of 10 Hz, and 20 MW cm^{-2} was the peak power output. The investigation confirmed that at innumerable stages of growth of barnacle larvae death of substantial proportion could be caused by a very low and smaller interval power pulsed irradiation. Two second duration of irradiation caused death of substantial proportion in the initial stages of nauplii larva. A considerable proportion opposition was observed concerning nauplii phase IV and cyprid larvae in comparison to larvae of phase II. The greater proportion of development of nauplii phase IV and cyprid larvae in comparison to larvae of phase II attributes the higher death rate of the latter. The physical damage of larvae due to irradiation was observed during microscopic analysis. The meddling of the cell membranes or protein or enzyme structures or anomalies in metabolism caused death due to irradiations by laser.

(e) Impact of localized, pulsed electric fields

Pérez et al. (2007) scrutinized the efficacy of pulsed electric fields (PEF) for fouling resistance application. The generation of low voltages (lesser than 25 V) high-strength electric fields dictates the usefulness of this methodology. The investigation setup comprised of titanium strips or electrodes placed in parallel orientation and reproduced via lithographic procedures onto a non-conducting material of borosilicate glass. The formation of an interdigitated electrode (IDE) structure was carried out by termination of each of the alternative electrode either at one of the twin mutual ends or collectors. It was observed that lower amplitude and shorter duration pulses were ineffective in comparison to pulses of amplitude of 10 V in and extent of 100 ms. The interdigitated electrode devices facilitate the generation of a localized, high-strength electric field. The amplification of electrochemical activity on the tips is due to the lesser distance between the adjoined electrode strips of opposite charge. Hence, the core technique for suppression of establishment might be due to intensity of field strength or electrochemical activity of both. The significance of voltage parameter was prominent when specifically 10 V displayed

repression of barnacle establishment to the extreme magnitude. This was assumed due to the effect of concentration gradients on the surroundings of the IDE strips which generated electro-osmotic flows. This electro-osmotic flows induced shear stresses repressed the establishment. Additionally, the phenomenon of electro-wetting created by pulsed electric fields lead to a water repelling surface which repressed the establishment of barnacles. This investigation emphasized that the suppression of establishment of barnacles was effective for certain duration of pulse amplitude of 10 V. The final observation was the pulses of 100 ms in duration, 200 Hz in frequency, and 10 V in pulse amplitude, averted establishment of barnacles lesser than 5%, whereas analogous IDE surfaces devoid of pulse application had an average establishment of barnacles of 40%.

(f) Influence of vibration frequency and amplitude

C.H. Choi et al. (2013) examined the association amid short frequency vibration (lesser than 500 Hz) and fouling discouragement in an in-situ investigation. For gauging the impacts of vibration on establishment of foulants acrylic panels with embedded lead zirconate titanate, known as PZT, were positioned in in-situ area for around 3 months and continuous vigorous excitement at diverse vibrational velocity levels of frequencies ranging between 70 and 445 Hz were applied. The vibrational wavelength if matches with the searching antennules then that particular vibrating surface was considered an unfavorable substrate. This led the cyprids to sway away and explore a substitute surface. This investigation also highlighted that hard foulants were the solitary fouling organisms that are influenced by the induced vibrational forces, and hence establishment on surfaces in considerably lesser numbers when specific frequencies and amplitudes are applied. The vibration of panels at comparatively upper frequencies (more than 260 Hz) unveiled the decrease in settlement of barnacles, whereas minor or nil effect was observed at lesser frequencies (70–100 Hz). This applied amplitude of vibration did not affect other fouling creatures such as ascidians, tubeworms, algae, and bryozoans. This investigation also revealed that an augment in the velocity amplitude of vibration was a prominent aspect in suppression of barnacle establishment.

Techniques to assess efficacy of barnacle resistant coatings

The selection and segregation of the investigating organisms for culture analysis in artificial surroundings is a decisive step and has to be ecologically relevant. The inclination to escalate the number of experimental species in laboratory analysis is to compare a widespread

representation of the activity spectra of an explicit formulation and as well as of its approach of action (Mokrini et al. 2008). The same formulation may be attractive, repulsive, or even toxic at different exposure levels, signifying the research prominence of varying proportions of concentrations. The artificial culture used for selection of newfangled fouling-resistant compositions and formulations or both could be segregated as antifouling analyses and toxicity evaluation (Dahms and Hellio 2009). The investigation of presence of fouling resistance nature can be executed over a widespread variety of foulants like fungi and bacteria (disc diffusion analysis using glass ring or paper disc techniques; spectrophotometric chemotaxis analyses; settlement-slide analyses; biochemical analyses), microalgae (suppression of growth, establishment, and adhesive nature), macroalgae (suppression of bonding of spores and zygotes and swaying nature of algal spore), and invertebrates (suppression of establishment of bryozoans and barnacles larvae, of mutation of polychaete larvae, suppression of attachment by juvenile mussels, and the synthesis of the phenoloxidase, activities of larvae) (Dahms and Hellio 2009). The toxic lab culture helps to identify acute, i.e., shorter contact of time, lesser than or equal to 96 h and chronic effects, i.e., contact of time varying from weeks to months, generally 10% of span of species of a compound or combination of compounds (Macken et al. 2009).

The coating substrate combined association comprises an extensive diversified assortment of materials, which starts from metals, alloys, polymers, and composites. The width of coating ranges from nano scale thinner films to micro scale in thickness. Both in and out of the globe vast majority of surfaces are being shielded by these coatings. Numerous internal as well as external forces result in adhesion failure of coatings. These majorly include machine-driven stress, surroundings persuaded stress of thermal origin, and surface oxidation and rusting. The assessment of bonding strength between applied surface and its coating is of precise importance, as the disaster of coatings is ordinarily a lengthy and sluggish progression for already pressed into service coating systems and therefore assessment of bonding strength can help to fast-track and dynamically stimulate the de-adhesion and delamination progression. For selection of effectual material blends for coating and surface combination and for the design and choice of improved coating techniques, the application of adhesion strength measurements can lay down methodologies. The combination of coatings and its surface can be characterized as ductile and ductile (D-D), brittle and brittle (B-B), brittle and ductile (B-D) systems, and vice versa. This grouping is based on the ability of a material to deform plastically. The materials which are ductile, prior onset of fracture demonstrates sizable plastic deformation with absorption of high energy,

whereas brittle materials undergo little or negligible plastic deformation with absorption of low energy. In the case of ductile materials, elongation of a crack is resisted by extensive plastic deformation unless the applied stress is increased; however, for a brittle material, rapid elongation of a crack occurs without increase of applied stress (Chen et al. 2019). After successful development of coatings, various assessment techniques for gauging the efficacy are also carried prior commercialization. These standardized techniques for measurement of parameters like viscosity, ultimate tensile strength, and Young's modulus are applicable for coatings. Rudawska and Jacniacka (2009) have highlighted the specific measurement technique for evaluation of water contact angle, sliding angle, and polar and dispersion components of surface free energy. For characterizing and determining the effect of stratification absorbance of both the top and bottom surface of the coatings, the technique called Fourier transform infrared spectroscopy (FTIR) can be implemented. The adhesion strength of barnacle-resistant coatings are measured by pseudo-barnacle tests (ASTMD 5618–2011) or pull-off analysis using Kendall's model (Singer et al. 2000). Apart from the abovementioned evaluation techniques, mechanically orientated micro-indentation testing, micro-scratch analysis (using both stylus and pulsed infrared laser techniques), induced laser shock spallation, and buckle and blister testing are few more methods implemented for understanding the efficacy of the fouling resistant coating. A thorough and detailed scrutiny of the comprehensive variety of journals has highlighted that appropriate adhesive nature must be demonstrated by the coatings on the applied region of the surfaces so that during their service lifetime designated benefits can be obtained. Notably for every coating and its applied substrate, a sole test is necessarily not resourceful or unanimously pertinent to inspect the adhesive nature. In various instances, to assess the efficacy of the coating, innumerable adhesion trials may be required. Hence, explicitly created coating substrate preparations might be required for some of the assessment techniques and this preparations might not be analogous to the real life preparation of coating (Chen et al. 2019).

Life cycle assessment

The recent emergence of stress on the idea of sustainability and biodegradability all across the globe is highly applauding. Though there are specific interpretations, complex consequences, and inclusion of numerous stakeholders, the majority concludes that sustainability necessitates creation of standards through stewardship of ecology, responsibility of society and growth of economy—both for present and forthcoming generations.

Deciding responsibly through well-adjusted selections and implementing resolutions that are best for ecosystem, societies, countries, businesses, and finally, the universe. Life cycle assessment (LCA) is one such yardstick that is important to develop an insight and encourage a more universal understanding of ecological benefits, curses, and trade-offs. The life cycle assessment approach is beneficial to decision-makers and stakeholders at all levels. LCA reports the possible ecological burdens and consumption of resources in each and every product life cycle step or during its service supply chain. This tool is primarily related with the estimation of the products' environmental impact, but of late this has been used for various purposes like building of cities, establishments, implementation of a project, implementation by industries, and many more (Gore and Kandasubramanian 2018; Kumar and Kandasubramanian 2019; Jain and Kandasubramanian 2020).

LCA methodology consist of four steps, namely goal and scope definition, inventory analysis, impact assessment, and interpretation (Fig. 12) which has been framed and outlined by the International Organization for Standardization (ISO) 14,040 series (ISO 14040:2006 2006).

- (a) Goal and scope definition: defining the intention of the LCA along with product systems, boundary conditions, audience, and functional unit are completed in this step. The geographical and temporal restrictions (depending on the lifetime of product and its emissions) about the product can be converged upon after a thorough understanding of the scope. Most interestingly, the functional unit is the key highlight of the LCA studies, and it is related with the consumption of the product. In this case, the functional unit can be the immersed

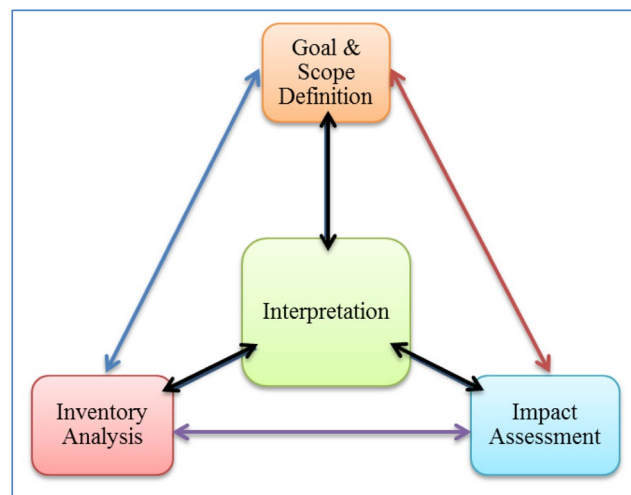


Fig. 12 Four steps of LCA

portion of the coated ship hull which offers protection from barnacle establishment. The functional unit can be considered as the service rendered by the coating for a specific period, e.g., years, and thus, can be assumed to be expressed as m^2/year .

- (b) Inventory analysis: determination of the environmental releases during each step, like resource extraction, material processing, is modeled in a flow chart. It is one of the evaluative steps of LCA and is guided by various related databases and also influenced by functional units. General inventory boundary description might be cradle to grave or cradle to gate approach.
- (c) Impact assessment: classification followed by quantification and measurement, followed by an analysis of the environmental impacts caused by the environmental releases, are determined in this step.
- (d) Interpretation: the outcome of the inventory analysis and impact assessment steps are discussed, conclusions are drawn on the potential impacts, and recommendations are made for further improvement in the whole process step and or usage step and or end of life step.

LCA reports concerning natural and synthetic additives in development of coatings are unavailable or might be in development process. A new life cycle assessment

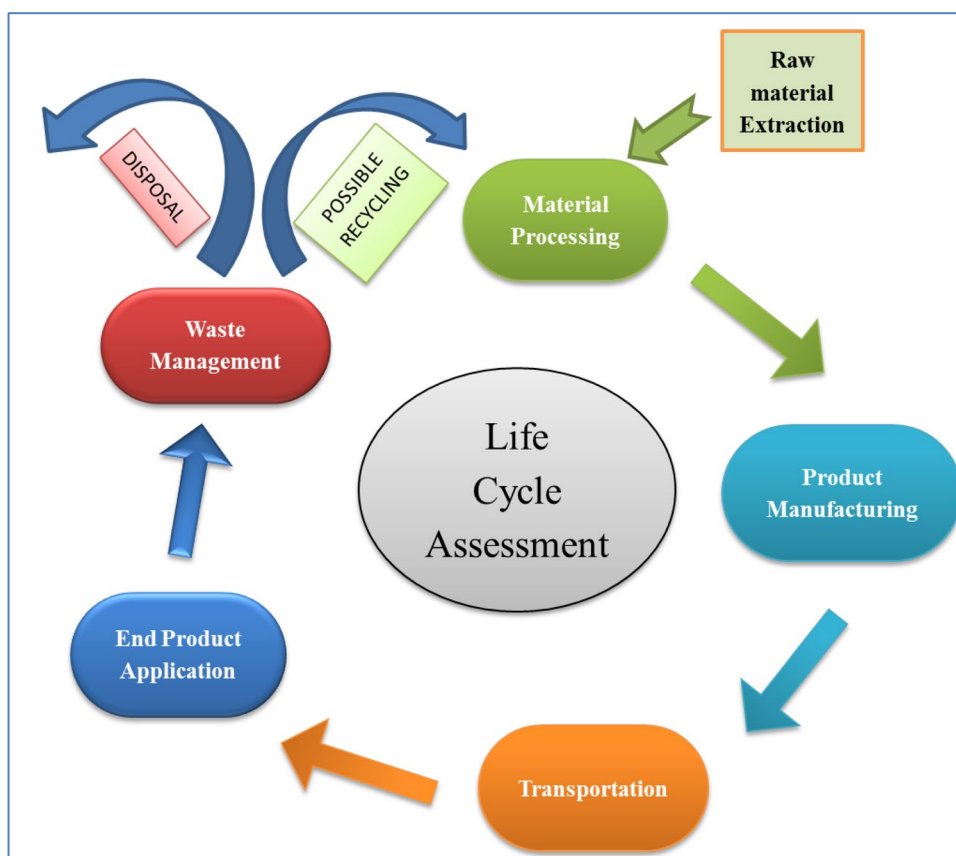
generally may range from months to years to be assessed and completed successfully. The reason behind this duration is because much information regarding inventory analysis and potential environmental impact has to be gathered, assessed, and interpreted (ISO 14040:2006 2006).

A typical LCA cycle of a natural product is highlighted (Fig. 13). The optimal usage of natural resources for raw materials followed by an energy-efficient production and systematic distribution via relevant modes of transportation and, subsequently, practical usage of the product (prolonged life of the end product reduces waste) are the initial vital steps for efficient resource optimization. Finally, minimization by 3Rs principle (i.e., Reduce, Recycle, and Reuse) or different disposal techniques (Fig. 14) are the final steps for structured waste management (Gore and Kandasubramanian 2018).

Environmental assessment

For carrying out the environmental assessment, three main indicators impact the assessment of global warming potential. The first indicator is global warming potential impact of the process, the second indicator is global warming potential effect of alternative products, and the third

Fig. 13 Typical LCA cycle of a natural product



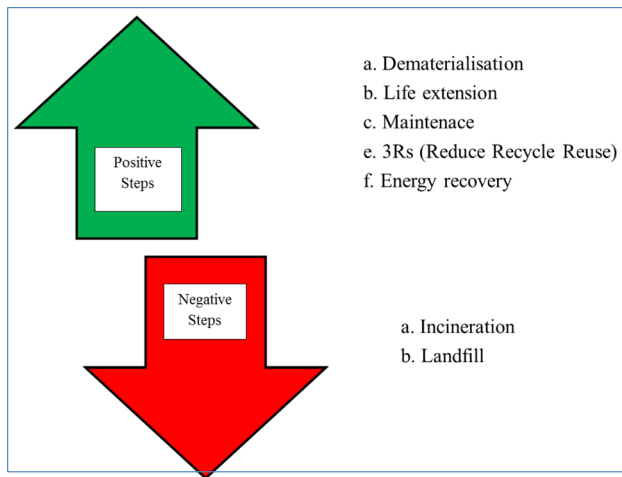


Fig. 14 Pros and cons in waste management

indicator is global warming potential total of the system (Mohandes and Salavati-Niasari 2014). The maximum values of global warming potential impact of the process are unveiled by thermal practices namely incineration, coin-cineration, and pyrolysis. It has been perceived that non-recovery techniques have greater global warming potential impact since the lowest recovery of products is recorded. This practice is more capital intensive because there is requirement of input materials like supercritical water, which enables greater return but, and simultaneously allows a reduction in global warming potential impacts. Hence, existence of a struggle between the environmental and economic dynamics is probable and till date of such recycling techniques is nonexistence, which entails both lesser global warming potential impact as well as is cheaper (Vo Dong et al. 2018).

Most of the natural additives constitute carbon and hydrogen as part of their configuration. This biogenic or non-biogenic carbon is either reused or recycled in another product cycle and can released as CH_4 or CO_2 during the waste management either due to combustion or decomposition, or is stockpiled as a consequence of waste management (due to landfilling or other actions that averts decomposition).

Calculation of carbon dioxide equivalent

The analytical expression for calculation of carbon dioxide equivalent (CO_2e) for an input, output, or a process when collection of data associated with the process or financial activity data is carried out is as follows (World Resources Institute 2020):

$$\text{CO}_2\text{e} = \text{Activitydata} \times \text{Emissionfactor} \times \text{GWP} \quad (1)$$

$$(\text{kg})(\text{unit})(\text{kgGHG}/\text{unit})(\text{kgCO}_2\text{e}/\text{kgGHG})$$

The analytical expression for calculation of inventory results for input, output, or process when direct emissions data had been collected and non-requirement of an emission factor is as follows:

$$\text{CO}_2\text{e} = \text{DirectEmissionsData} \times \text{GWP} \quad (2)$$

$$(\text{kg})(\text{kgGHG})(\text{kgCO}_2\text{e}/\text{kgGHG})$$

where CO_2e refers to carbon dioxide equivalent, GWP refer to global warming potential, and GHG refers to greenhouse gases.

Recent carbon footprint data by global paint companies

Greenhouse gases emissions are categorized under three scopes which are the following:

- Scope 1 standard represents company owned or regulated direct emissions from sources. This includes onsite combustion units, heat exchangers.
- Scope 2 standard represents company owned or regulated indirect emissions from sources. This includes electricity, steam, heat, and cooling from service providers.
- Scope 3 standard emissions result from the sources that are not owned or directly controlled by a company but associated to its activities. This includes activities such as employee travel, shipping, procurement, product use, and waste disposal.

The Dutch multinational company known as AkzoNobel (2019) produces performance coatings and paints for both business and clients internationally, highlighted in their sustainability report of 2019 that since 2015 to 2019 they has drastically reduced and regulated its carbon footprint. Few highlights are:

- Direct CO_2e emissions (scope 1) was reduced from 24.47 to 18.18 (kg/ton) from 2015 to 2019 correspondingly.
- Indirect CO_2e emissions (scope 2) was reduced from 78.73 to 57.13 (kg/ton) from 2015 to 2019 correspondingly.
- Cradle-to-grave carbon footprint (scope 1, 2, and 3) was reduced from 15.0 to 14.6 million tons from 2018 to 2019 correspondingly.

The Norwegian chemicals company known as Jotun Group is dealing chiefly in decorative paints and performance coatings, highlighted in their sustainability report of 2019 that since 2015 to 2019 it has reduced emissions per tonne from 90 to 77 kg CO_2 per tonne (Jotun Group 2021).

PPG Industries UK Ltd (2019) is an American Fortune 500 company and worldwide seller of coatings, paints, and specialty materials, highlighted in their sustainability report of 2019. Since 2015 to 2019, they have been reducing the emissions of greenhouse gas primarily by lessening their consumption of energy. Few highlights are:

- (a) Direct CO₂ (e) emissions was reduced from 1.04 to 0.38 (million metric tons of carbon dioxide equivalents) from 2015 to 2019 correspondingly.
- (b) Indirect CO₂ (e) emissions was reduced from 0.98 to 0.50 ((million metric tons of carbon dioxide equivalents) from 2015 to 2019 correspondingly.

Waste management

Recycling is the capability of a material to preserve its originals state characteristics. However, numerous constituents cannot endure recycling due to their failure to maintain their default properties even after exhaustion of its serviceable life. Usage of petrochemicals as preparatory constituents is detrimental as they are non-renewable. Recycling index (R) is a parameter for comparing materials, which is the ratio of the post-recycling cost of the material against the minimal possible cost of the material (Awais et al. 2015). The analytical expression is:

$$R = V_p/V_m \quad (3)$$

where the lowermost probable rate of a material (rupee/kg) is referred to as V_m and is the post-recycling rate of the material (rupee/kg) is referred to as V_p . The material is found to be recyclable if the value of R approaches to unity.

Both mechanical recyclability and chemical recyclability restores the robustness of the elements back to a superior value. However, it is a lesser primacy decision, primarily due to a greater proportion of material and energy inputs than the preceding life extension substitutes. Lastly, in feedstock, the polymeric constituent's recyclability is reclaimed into fuel-based products, and recovery of energy is to be considered as lesser-value alternatives. However, it is understood that recycling or recovery substitute, or a grouping of these, would provide the improved ecologically responsive value which has to be assessed.

Future research areas and conclusion

Prohibition of tributyltin and further prohibition of copper-based biocides encouraged the search, identification, and development of eco-friendly, biodegradable, non-toxic compounds to inhibit barnacle establishment substrates. For

the development of anti-barnacle coating, it is understood that either the substrate's surface can be modified to prevent establishment of barnacles or coating can be applied made from natural or synthetic additives is fatal to cypris larvae or barnacles either by leaching, biocide release, or contact killing mechanism. After vigorous, detailed, and thorough scrutiny of recent review and research articles on anti-barnacle compounds, the following way ahead or focus areas are collated below:

- (a) The majority of the research for the expansion of ecologically responsive anti-barnacle coating was restricted to *Balanus amphitrite* or *Balanus albicostatus* for experimental analysis. The search and identification of more barnacle species should be undertaken as well as immediate attention on further field tests of all these barnacle species with the help of these known natural extracts (isolated from marine and or territorial plants and organisms) should be focused. The recommended natural extracts are tabulated (Table 1) based on their benchmark parameters:
- (b) A detailed study of the adhesion mechanism, as well as the composition of the barnacle cement, needs to be also researched. Few articles are available in this regard are applauded; however, detailed analysis in a structured manner is utterly essential for further progress in developing novel anti barnacle strategies and products.
- (c) The inference from all the related review and research papers highlights that the outcome of laboratory analyses do not provide adequate information concerning the field performance of the coatings or compounds. Many experiments conducted in the laboratory deployed cyprids larvae or barnacles cultured in laboratories or artificial atmosphere. As a result of which the efficacy of the novel compounds varies significantly from laboratory experiments to field tests. The experimental analysis with requisite supporting documents should be made available which would be providing a key impetus in the development of anti-barnacle coatings.
- (d) Maritime biological fouling is the outcome of a widespread array of biotic processes that involve physiological and chemical dynamics, and replicating analogous multifarious occurrences in a workshop environment is quite challenging. Nevertheless, requirement of culture-dish tests are essential, as the straight forward assessment of antifouling coatings in real ground setup is costly (secured location, paints and panels, secured rafts accessibility and maintenance of raft, volunteers to execute a systematic macro-fouling scrutiny of submerged surfaces with coats of compositions) and

Table 1 List of recommended ecologically benign natural extracts and their chief parameters

Name	Source	Target species	Parameters	Chemical structure (Appendix)	Ref
Avarol (sesquiterpenoid hydroquinone)	Mediterranean sponge <i>Dysidea avara</i>	Cyprids of the barnacle <i>Balanus amphitrite</i>	ED ₅₀ 0.65 µg/mL LD ₅₀ 13.3 µg/mL (low toxicity)	1	(Tsoukatou et al. 2007)
Avarone (sesquiterpenoid quinone)	Mediterranean sponge <i>Dysidea avara</i>	Cyprids of the barnacle <i>Balanus amphitrite</i>	ED ₅₀ 3.41 µg/mL, LD ₅₀ 27.2 µg/mL (low toxicity)	2	(Tsoukatou et al. 2007)
3 ^l -(p-chlorophenyl) avarone	Mediterranean sponge <i>Dysidea avara</i>	Cyprids of barnacle	ED ₅₀ 0.65 µg/mL LD ₅₀ > 50 µg/mL	3	(Qi et al. 2008b)
4 ^l -Propylthioavarone	Mediterranean sponge <i>Dysidea avara</i>	Cyprids of barnacle	ED ₅₀ 0.45 mg/mL LD ₅₀ > 50 µg/mL	4	(Qi et al. 2008b)
Subergorgic acid	South China Sea gorgonian <i>Subergorgia suberosa</i>	Barnacle (<i>Balanus amphitrite</i>) Bryozoan (<i>Bugula neritina</i>)	EC ₅₀ 1.2 µg/mL EC ₅₀ 3.2 µg/mL	5	(Qi et al. 2008b)
Primarene diterpene	Mangrove plant <i>Cerriops tagal</i>	Barnacle <i>Balanus albicostatus</i>	ED ₅₀ 0.04 µg/mL, LD ₅₀ > 10 µg/cm ² low toxicity	6	(Chen et al. 2008)
Four variants of briarane diterpenes	South China Sea gorgonian <i>Junceella juncea</i>	Cyprids of <i>Balanus amphitrite</i>	7: No Data 8: ED ₅₀ 0.004 µg/mL 9: ED ₅₀ 0.005 µg/mL 10: ED ₅₀ 22.6 µg/mL	7,8,9,10	(Qi et al. 2006, 2009)
Three variants of meroditerpenes	Mediterranean brown alga <i>Halidrys siliquosa</i>	11 & 12 compounds were highly toxic to naupliar larvae 13: No activity	11: LD ₅₀ 5 µg/mL 12: LD ₅₀ 10 µg/mL 13 was non-toxic even at 100 µg/mL	11,12,13	(Culioli et al. 2008)
Hydroquinone C acetate (a terpenoids variant)	Mediterranean sponges	Toxic against barnacle larvae	ED ₅₀ < 1 µg/mL LD ₅₀ > 100 µg/mL	14	(Hellio et al. 2005)
Dihydrofurospingin II (a terpenoids variant)	Mediterranean sponges	Toxic against barnacle larvae	ED ₅₀ < 2.5 µg/mL LD ₅₀ > 100 µg/mL	15	(Hellio et al. 2005)
Compound 17 is A 5α-hydroxylated analog of steroid 16	Gorgonian <i>Subergorgia suberosa</i> of South China Sea	Antifouling nature against <i>Bugula neritina</i> and <i>Balanus amphitrite</i>	–	16, 17	(Qi et al. 2010)
Novel A-nor steroid	Marine sponge <i>Acanthella cavernosa</i>	Barnacle species <i>Balanus albicostatus</i>	EC ₅₀ 8–32 µg/mL	18	(Qiu et al. 2008)
Aureobasidin	Marine fungus of the genus <i>Aureobasidium</i>	Inhibited larval establishment of <i>Balanus amphitrite</i>	–	19	(Abdel-Lateff et al. 2009)
Barettin (brominated diketopiperazine)	Marine sponge <i>Geodia baretti</i> of Swedish origin	Antifouling against the barnacle <i>Balanus improvises</i>	EC ₅₀ 0.9 µM	26	(Sjögren et al. 2004b)
8,9-Dihydrobarettin (brominated diketopiperazines)	Marine sponge <i>Geodia baretti</i> of Swedish origin	Antifouling against the barnacle <i>Balanus improvises</i>	EC ₅₀ 7.8 µM	27	(Johnson et al. 2004)
3-Chloro-2,5-dihydroxybenzyl alcohol	The metabolic product of aquatic fungus <i>Ampelmyces</i> sp.	Repressed larval establishment of <i>Balanus amphitrite</i>	EC ₅₀ 3.19–3.81 µg/mL LC ₅₀ 266.7 µg/mL	30	(Kwong et al. 2006)
Luteolin-4 ^l -glucuronide	A flavonoid glycoside derived the seagrass <i>Enhalus acoroides</i>	Subdued larval establishment of <i>Balanus amphitrite</i>	EC ₅₀ 0.52 µg/mL	31	(Qi et al. 2008a)
Alkylated butenolides	Actinomycete <i>Streptomyces</i> sp. found in deep sea sediments	Antifouling against larvae of <i>Balanus amphitrite</i>	EC ₅₀ 8–15 µg/mL LD ₅₀ > 100 µg/mL	33, 34, 35	(Xu et al. 2010)
Floridoside	Extracted from the red alga <i>Grateloupia turuturu</i>	Exhibited effective anti-barnacle activity	EC ₅₀ < 1 µg/mL	38	(Hellio et al. 2004)

Table 1 (continued)

Name	Source	Target species	Parameters	Chemical structure (Appendix)	Ref
MANOALIDE	Marine sponge <i>Luffariella variabilis</i>	Inhibit <i>Balanus amphitrite</i> larvae	EC ₅₀ 0.24–0.80 µg/mL	40	(Tsukamoto et al. 1997)
Zosteric acid or p-(sulfoxy) cinnamic acid	Seagrass <i>Zostera marina</i>	Anti-barnacle activity	(EC ₅₀ ~ 170–450 ppm)	44	165–173
2'-Methoxykurarinone	Plant <i>Sophora flavescens</i> of medicinal properties	Subdued establishment of cyprids of <i>Balanus albicostatus</i>	EC ₅₀ 2.0 µg/mL LC ₅₀ > 25 µg/mL (low toxicity)	48	(Feng et al. 2009b)
Matrine	Plant <i>Sophora flavescens</i> of medicinal properties	Surpassed establishment of cyprids of <i>Balanus albicostatus</i>	EC ₅₀ 7.1 µg/mL, LC ₅₀ > 250 µg/mL (low toxicity)	49	(Feng et al. 2009b)

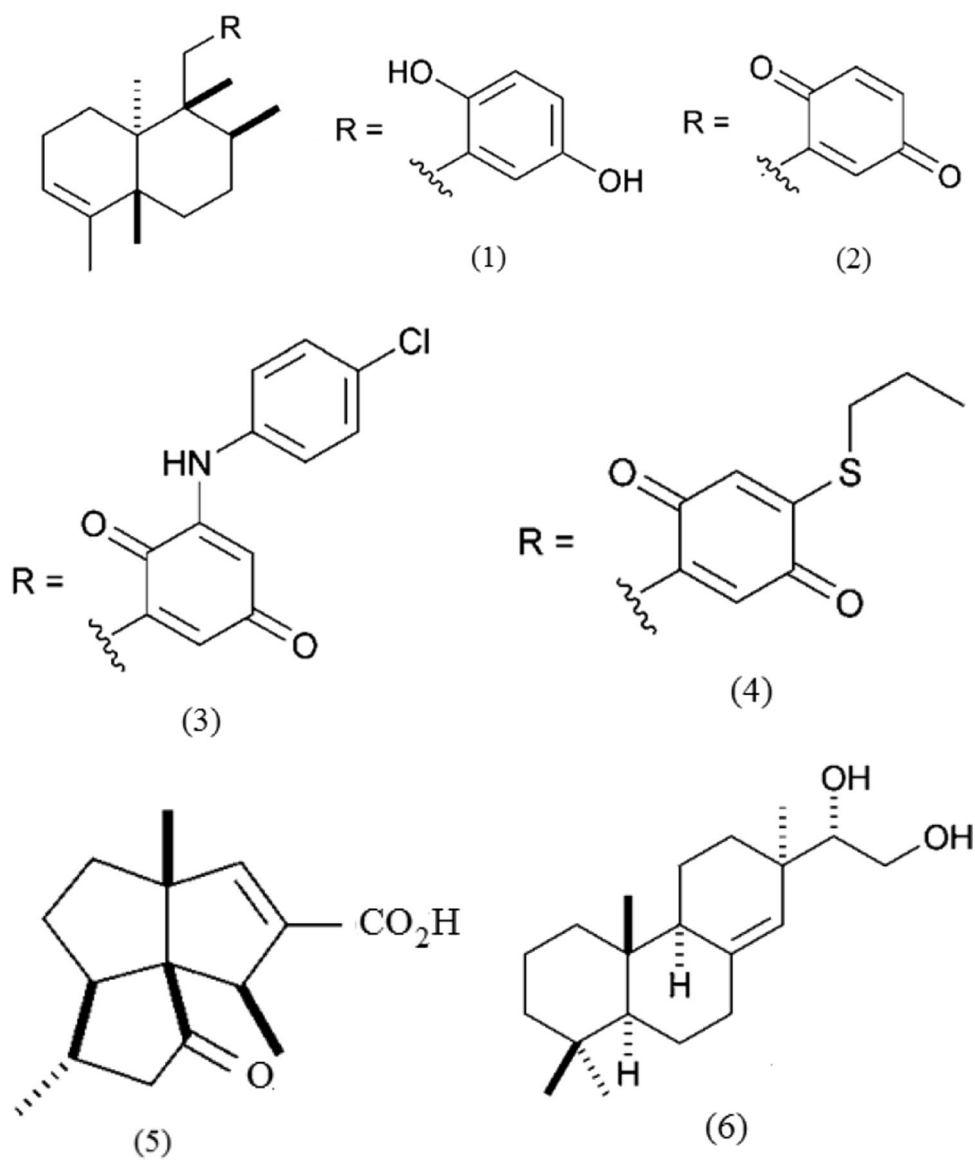
requirement of several months to years indicative of prospective analysis period. There is a common understanding that, because of overall biodiversity in aquatic biomes, a sole element might not impede the establishment and evolution of all the creatures associated in this aquatic biological fouling process. Accordingly, numerous diverse entities of specific origin are prerequisite for assessing the antifouling possibility of biocides and coatings.

- (e) The extracts from the marine sponges and invertebrates and terrestrial plants that all showed the anti-barnacle property should be available naturally in abundance; otherwise, successful commercial production might not be possible. This implies the development of eco-friendly products needs an essential pre-scan regarding the optimal availability of raw material prior to initiation of product development.
- (f) Biological mimicking or biomimetics, which refers to imitating nature's designs, is the latest advancement in antifouling strategies. The prominence of surface topologies and the interaction influencing antifouling qualities is addressed by biomimetics, where the aim is to comprehend and acquire modules from nature and transform these models for man-made applications. Focused attention is imparted to naturally occurring structures, materials, and surface. These learning's are transformed for development and further application on a commercial scale. Examples like the hydrophobicity of the lotus leaves, low-drag shape of dolphins, adhesive gecko feet, riblet micro texture of shark's skin, oil-resistant fish skin, have been successfully studied and applied for human-made usage purposes. However, the same properties can be modified for

developing anti barnacle strategies and or products. (Telegdi et al. 2016).

- (g) The dirt of specific LCA data on natural and synthetic coatings have surfaced while carrying out the related literature survey. Different modeling methods and software are available which can be implemented for analysis for better understanding the importance of LCA.
- (h) Life cycle analysis may be either comparison perspective or solitary (i.e., an analysis of a single product) study. A solitary study establishes the period (invention, usage, or disposition) that results in maximum influence. A comparison perspective study highlights the ecologically desirable substitute. Throughout the life cycle approach, the framework might encompass from cradle to gate, cradle to grave, or cradle to cradle methodologies. Cradle to cradle specifies procedures for disposing a product and reverting to the biodiversity; the life stages of natural coatings can be considered analogous because during the degradation of poly lactic acid the carbonaceous element is recycled back into the ecology for biomass absorption. In a life cycle assessment of polymers, boundaries may be considered a cradle to pellet methodology; where the production of pellets of resin marks the end of assessment, and sequential methodologies include end product manufacturing, usage, and discarding are omitted from assessment steps.
- (i) For development and research of capable barnacle-resistant coatings, numerous preliminary models of roughness function have to be developed by scrutinizing several types of fouling situations and various sizes of barnacles.

Appendix



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References

- Abdel-Lateff A, Elkhayat ES, Fouad MA, Okino T (2009) Aureobasidin, new antifouling metabolite from marine-derived fungus *Aureobasidium* sp. *Nat Prod Commun* 4:389–394. <https://doi.org/10.1177/1934578x0900400315>
- AkzoNobel (2019) Environmental - AkzoNobel Report 2019
- Aldred N, Phang IY, Conlan SL et al (2008) The effects of a serine protease, Alcalase, on the adhesives of barnacle cyprids (*Balanus amphitrite*). *Biofouling* 24:97–107. <https://doi.org/10.1080/08927010801885908>
- Almeida JR, Correia-Da-Silva M, Sousa E et al (2017) Antifouling potential of nature-inspired sulfated compounds. *Sci Rep* 7:1–11. <https://doi.org/10.1038/srep42424>
- Almeida JR, Vasconcelos V (2015) Natural antifouling compounds: effectiveness in preventing invertebrate settlement and adhesion. *Biotechnol Adv* 33:343–357. <https://doi.org/10.1016/j.biotechadv.2015.01.013>
- Alzieu C (2000) Impact of Tributyltin on marine invertebrates. *Eco-toxicology* 9:71–76. <https://doi.org/10.1023/A:1008968229409>
- Amandi R, Hyde JR, Ross SK et al (2005) Continuous reactions in supercritical fluids; a cleaner, more selective synthesis of thymol in supercritical CO₂. *Green Chem* 7:288–293. <https://doi.org/10.1039/b418983c>
- Anil AC, Khandeparker L, Desai DV et al (2010) Larval development, sensory mechanisms and physiological adaptations in acorn barnacles with special reference to *Balanus amphitrite*. *J Exp Mar Bio Ecol* 392:89–98. <https://doi.org/10.1016/j.jembe.2010.04.012>
- Awais M, Tausif M, Ahmad F et al (2015) Inclusion of recycled PPTA fibre in development of cut-resistant gloves. *J Text Inst* 106:354–358. <https://doi.org/10.1080/00405000.2014.922246>
- Ba M, Zhang ping Z, hong Qi Y (2019) The influence of MWCNTs-OH on the properties of the fouling release coatings based on polydimethylsiloxane with the incorporation of phenylmethylsilicone oil. *Prog Org Coatings* 130:132–143. <https://doi.org/10.1016/j.porgcoat.2019.01.055>
- Bangar GY, Ghule D, Singh RKP, Kandasubramanian B (2018a) Overview of failure modes of pseudo compositionally graded thermal barrier coatings and remedies for augmenting the service life of the coated components. *Mater Focus* 7:741–750. <https://doi.org/10.1166/mat.2018.1590>
- Bangar GY, Ghule D, Singh RKP, Kandasubramanian B (2018b) Thermally triggered transition of fluid atomized micro- and nanotextured multiscale rough surfaces. *Colloids Surfaces A Physicochem Eng Asp* 549:212–220. <https://doi.org/10.1016/j.colsurfa.2018.03.044>
- Barrios CA, Xu Q, Cutright T, Newby BMZ (2005) Incorporating zosteric acid into silicone coatings to achieve its slow release while reducing fresh water bacterial attachment. *Colloids Surfaces B Biointerfaces* 41:83–93. <https://doi.org/10.1016/j.colsurfb.2004.09.009>
- Bergamo VZ, Balbuena EA, Hatwig C et al (2015) 1-n-Hexadecyl-3-methylimidazolium methanesulfonate and chloride salts with effective activities against *Candida tropicalis* biofilms. *Lett Appl Microbiol* 61:504–510. <https://doi.org/10.1111/lam.12488>
- Bielecki J, Chan BKK, Hoeg JT, Sari A (2009) Antennular sensory organs in cyprids of balanomorphan cirripedes: standardizing terminology using *Megabalanus rosa*. *Biofouling* 25:203–214. <https://doi.org/10.1080/08927010802688087>
- Blanck H, Eriksson KM, Grönvall F et al (2009) A retrospective analysis of contamination and periphyton PICT patterns for the antifoulant irgarol 1051, around a small marina on the Swedish west coast. *Mar Pollut Bull* 58:230–237. <https://doi.org/10.1016/j.marpolbul.2008.09.021>
- Braga PC, Alfieri M, Culici M, Dal Sasso M (2007) Inhibitory activity of thymol against the formation and viability of *Candida albicans* hyphae. *Mycoses* 50:502–506. <https://doi.org/10.1111/j.1439-0507.2007.01412.x>
- Brown GD, Wong HF, Hutchinson N et al (2004) Chemistry and biology of macrolactone A from the marine cyanobacterium *Kryptothrix maculans*. *Phytochem Rev* 3:381–400. <https://doi.org/10.1007/s11101-004-6552-5>
- Callow ME, Callow JA (2002) Marine biofouling: a sticky problem. *Biologist* 49:10–14
- Callow ME, Callow JA (1998) Attachment of zoospores of the fouling alga *Enteromorpha* in the presence of zosteric acid. *Biofouling* 13:87–95. <https://doi.org/10.1080/08927019809378373>
- Candries M, Anderson CD, Atlar M (2001) Foul release systems and drag. Consolidation of technical advances in the protective and marine coatings industry. *Proc PCE 2001 Conf Antwerp* 273–286
- Cattò C, Dell’Orto S, Villa F et al (2015) Unravelling the structural and molecular basis responsible for the anti-biofilm activity of zosteric acid. *PLoS ONE* 10:1–24. <https://doi.org/10.1371/journal.pone.0131519>
- Champ MA (2000) A review of organotin regulatory strategies, pending actions, related costs and benefits. *Sci Total Environ* 258:21–71. [https://doi.org/10.1016/S0048-9697\(00\)00506-4](https://doi.org/10.1016/S0048-9697(00)00506-4)
- De CJ, Qiu Y, Yang ZW et al (2008) Dimeric diterpenes from the roots of the mangrove plant *Ceriops tagal*. *Helv Chim Acta* 91:2292–2298. <https://doi.org/10.1002/hlca.200890249>
- Chen L, Qian PY (2017) Review on molecular mechanisms of antifouling compounds: an update since 2012. *Mar Drugs* 15:1–20. <https://doi.org/10.3390/md15090264>
- Chen X, Shaw C, Gelman L, Grattan KTV (2019) Advances in test and measurement of the interface adhesion and bond strengths in coating-substrate systems, emphasising blister and bulk techniques. *Measurement* 139:387–402. <https://doi.org/10.1016/j.measurement.2019.03.026>

- Chesworth JC, Donkin ME, Brown MT (2004) The interactive effects of the antifouling herbicides Irgarol 1051 and Diuron on the seagrass *Zostera marina* (L.). *Aquat Toxicol* 66:293–305. <https://doi.org/10.1016/j.aquatox.2003.10.002>
- Choi CH, Scardino AJ, Dylejko PG et al (2013) The effect of vibration frequency and amplitude on biofouling deterrence. *Biofouling* 29:195–202. <https://doi.org/10.1080/08927014.2012.760125>
- Cima F, Ballarin L (2012) Immunotoxicity in ascidians: antifouling compounds alternative to organotin. III - The case of copper(I) and Irgarol 1051. *Chemosphere* 89:19–29. <https://doi.org/10.1016/j.chemosphere.2012.04.007>
- Cimanga K, Apers S, De Bruyne T et al (2002) Chemical composition and antifungal activity of essential oils of some aromatic medicinal plants growing in the Democratic Republic of Congo. *J Essent Oil Res* 14:382–387. <https://doi.org/10.1080/10412905.2002.9699894>
- Culioli G, Ortalo-Magné A, Valls R et al (2008) Antifouling activity of meroditerpenoids from the marine brown alga *halidrys siliquosa*. *J Nat Prod* 71:1121–1126. <https://doi.org/10.1021/np070110k>
- Dafforn KA, Lewis JA, Johnston EL (2011) Antifouling strategies: history and regulation, ecological impacts and mitigation. *Mar Pollut Bull* 62:453–465. <https://doi.org/10.1016/j.marpolbul.2011.01.012>
- Dahms HU, Hellio C (2009) Laboratory bioassays for screening marine antifouling compounds. In: *advances in marine antifouling coatings and technologies*. Elsevier, pp 275–307
- Dalleau S, Cateau E, Bergès T et al (2008) In vitro activity of terpenes against *Candida* biofilms. *Int J Antimicrob Agents* 31:572–576. <https://doi.org/10.1016/j.ijantimicag.2008.01.028>
- Dang and Lovellc (2015) Reviews-2015-Dang-91.full. *Am Soc Microbiol* 80:91–138. <https://doi.org/10.1128/MMBR.00037-15.Address>
- Davidson IC, Brown CW, Sytsma MD, Ruiz GM (2009) The role of containerships as transfer mechanisms of marine biofouling species. *Biofouling* 25:645–655. <https://doi.org/10.1080/08927010903046268>
- Demirel YK, Uzun D, Zhang Y et al (2017) Effect of barnacle fouling on ship resistance and powering. *Biofouling* 33:819–834. <https://doi.org/10.1080/08927014.2017.1373279>
- Dexter SC (1979) Influence of substratum critical surface tension on bacterial adhesion-in Situ studies. *J Colloid Interface Sci* 70:346–354. [https://doi.org/10.1016/0021-9797\(79\)90038-9](https://doi.org/10.1016/0021-9797(79)90038-9)
- Dexter SC, Sullivan JD, Williams J, Watson SW (1975) Influence of substrate wettability on the attachment of marine bacteria to various surfaces. *Appl Microbiol* 30:298–308. <https://doi.org/10.1128/aem.30.2.298-308.1975>
- Dobretsov S, Xiong H, Xu Y et al (2007) Novel antifoulants: inhibition of larval attachment by proteases. *Mar Biotechnol* 9:388–397. <https://doi.org/10.1007/s10126-007-7091-z>
- Doria OF, Castro R, Gutierrez M, et al (2018) Novel alkylimidazolium ionic liquids as an antibacterial alternative to pathogens of the skin and soft tissue infections. *Molecules* 23. <https://doi.org/10.3390/molecules23092354>
- Egorova KS, Ananikov VP (2018) Fundamental importance of ionic interactions in the liquid phase: A review of recent studies of ionic liquids in biomedical and pharmaceutical applications. *J Mol Liq* 272:271–300. <https://doi.org/10.1016/j.molliq.2018.09.025>
- Faÿ F, Linossier I, Langlois V et al (2006) Degradation and controlled release behavior ϵ -caprolactone copolymers in biodegradable antifouling coatings. *Biomacromol* 7:851–857. <https://doi.org/10.1021/bm0509669>
- Faÿ F, Linossier I, Langlois V, Vallée-Rehel K (2007a) Biodegradable poly(ester-anhydride) for new antifouling coating. *Biomacromol* 8:1751–1758. <https://doi.org/10.1021/bm061013t>
- Faÿ F, Linossier I, Legendre G, Vallée-Rehel K (2008) Micro-encapsulation and antifouling coatings: development of poly(lactic acid) microspheres containing bioactive molecules. *Macromol Symp* 272:45–51. <https://doi.org/10.1002/masy.200851205>
- Faÿ F, Renard E, Langlois V et al (2007b) Development of poly(ϵ -caprolactone-co-l-lactide) and poly(ϵ -caprolactone-co- δ -valerolactone) as new degradable binder used for antifouling paint. *Eur Polym J* 43:4800–4813. <https://doi.org/10.1016/j.eurpolymj.2007.07.045>
- Feng D, Ke C, Li S et al (2009a) Pyrethroids as promising marine antifoulants: laboratory and field studies. *Mar Biotechnol* 11:153–160. <https://doi.org/10.1007/s10126-008-9130-9>
- Feng DQ, Ke CH, Lu CY, Li SJ (2009b) Herbal plants as a promising source of natural antifoulants: evidence from barnacle settlement inhibition. *Biofouling* 25:181–190. <https://doi.org/10.1080/08927010802669210>
- Fletcher M, Marshall KC (1982) Are solid surfaces of ecological significance to aquatic bacteria? 199–236. https://doi.org/10.1007/978-1-4615-8318-9_6
- Fletcher RL, Callow ME (1992) The settlement, attachment and establishment of marine algal spores. *Br Phycol J* 27:303–329. <https://doi.org/10.1080/00071619200650281>
- Fu Z, Yu Y, Yin D et al (2005) Vapor-phase highly selective O-methylation of catechol with methanol over ZnCl₂ modified γ -Al₂O₃ catalysts. *J Mol Catal A Chem* 232:69–75. <https://doi.org/10.1016/j.molcata.2005.01.030>
- Galhenage TP, Hoffman D, Silbert SD et al (2016) Fouling-release performance of silicone oil-modified siloxane-polyurethane coatings. *ACS Appl Mater Interfaces* 8:29025–29036. <https://doi.org/10.1021/acsami.6b09484>
- Gallo A, Tosti E (2015) Reprotoxicity of the antifoulant chlorothalonil in ascidians: An ecological risk assessment. *PLoS ONE* 10:1–14. <https://doi.org/10.1371/journal.pone.0123074>
- Gharde S, Kandasubramanian B (2018) The importance of electrodeless metallic build-up on surface modified substrates for multifunctional engineering applications: a recent progress update. *Trans Indian Inst Met* 71:2873–2892. <https://doi.org/10.1007/s12666-018-1397-6>
- Gore PM, Balakrishnan S, Kandasubramanian B (2019a) Superhydrophobic corrosion inhibition polymer coatings. *Superhydrophobic Polym Coatings* 223–243. <https://doi.org/10.1016/B978-0-12-816671-0.00011-4>
- Gore PM, Gawali P, Naebe M, et al (2020a) Polycarbonate and activated charcoal-engineered electrospun nanofibers for selective recovery of oil/solvent from oily wastewater. *SN Appl Sci* 2. <https://doi.org/10.1007/s42452-020-03609-x>
- Gore PM, Kandasubramanian B (2018) Functionalized aramid fibers and composites for protective applications: a review. *Ind Eng Chem Res* 57:16537–16563. <https://doi.org/10.1021/acs.iecr.8b04903>
- Gore PM, Naebe M, Wang X, Kandasubramanian B (2020b) Silk fibres exhibiting biodegradability & superhydrophobicity for recovery of petroleum oils from oily wastewater. *J Hazard Mater* 389. <https://doi.org/10.1016/j.jhazmat.2019.121823>
- Gore PM, Naebe M, Wang X, Kandasubramanian B (2021) Nanofluoro dispersion functionalized superhydrophobic degummed & waste silk fabric for sustained recovery of petroleum oils & organic solvents from wastewater. *J Hazard Mater* 127822. <https://doi.org/10.1016/j.jhazmat.2021.127822>
- Gore PM, Purushothaman A, Naebe M, et al (2019b) Nanotechnology for oil-water separation. 299–339. https://doi.org/10.1007/978-3-030-02381-2_14
- Gottenbos B, Van Der Mei HC, Klatter F et al (2002) In vitro and in vivo antimicrobial activity of covalently coupled quaternary ammonium silane coatings on silicone rubber. *Biomaterials* 23:1417–1423. [https://doi.org/10.1016/S0142-9612\(01\)00263-0](https://doi.org/10.1016/S0142-9612(01)00263-0)
- Grapski JA, Cooper SL (2001) Synthesis and characterization of non-leaching biocidal polyurethanes. *Biomaterials* 22:2239–2246. [https://doi.org/10.1016/S0142-9612\(00\)00412-9](https://doi.org/10.1016/S0142-9612(00)00412-9)

- Guardiola FA, Cuesta A, Meseguer J, Esteban MA (2012) Risks of using antifouling biocides in aquaculture. *Int J Mol Sci* 13:1541–1560. <https://doi.org/10.3390/ijms13021541>
- Guo S, Khoo BC, Teo SLM, Lee HP (2013) The effect of cavitation bubbles on the removal of juvenile barnacles. *Colloids Surfaces B Biointerfaces* 109:219–227. <https://doi.org/10.1016/j.colsurfb.2013.03.046>
- Guo S, Lee HP, Khoo BC (2011) Inhibitory effect of ultrasound on barnacle (*Amphibalanus amphitrite*) cyprid settlement. *J Exp Mar Bio Ecol* 409:253–258. <https://doi.org/10.1016/j.jembe.2011.09.006>
- Hallers-tjabbes CC, Boon JP (n.d.) Communicating the harmful impact of TBT: what can Scientists Contribute to EU Environmental Policy Planning in a Global Context? CaTO Marine Ecosystems, The Netherlands Superior, Huelva, Spain University of Nottingham, U
- Hany R, Bo C, Geiger T, et al (2004) Toward non-toxic antifouling : synthesis of hydroxy-, poly [3-hydroxyalkanoates]. *Biomacromolecules* 1452–1456
- Hayase N, Sogabe T, Itou R et al (2003) Polymer film produced by a marine bacterium. *J Biosci Bioeng* 95:72–76. [https://doi.org/10.1016/S1389-1723\(03\)80151-2](https://doi.org/10.1016/S1389-1723(03)80151-2)
- Hellio C, Maréchal JP, Da Gama BAP, et al (2009) Natural marine products with antifouling activities. *Adv Mar Antifouling Coatings Technol* 572–622 <https://doi.org/10.1533/9781845696313.3.572>
- Hellio C, Simon-Colin C, Clare AS, Deslandes E (2004) Isethionic acid and floridose isolated from the red alga, *Grateloupia turuturu*, inhibit settlement of *Balanus amphitrite* cyprid larvae. *Biofouling* 20:139–145. <https://doi.org/10.1080/08927010412331279605>
- Hellio C, Tsoukatou M, Maréchal JP et al (2005) Inhibitory effects of Mediterranean sponge extracts and metabolites on larval settlement of the barnacle *Balanus amphitrite*. *Mar Biotechnol* 7:297–305. <https://doi.org/10.1007/s10126-004-3150-x>
- Howell D, Behrends B (2006) A review of surface roughness in antifouling coatings illustrating the importance of cutoff length. *Biofouling* 22:401–410. <https://doi.org/10.1080/08927010601035738>
- Hunsucker KZ, Braga C, Gardner H et al (2019) Using ultraviolet light for improved antifouling performance on ship hull coatings. *Biofouling* 35:658–668. <https://doi.org/10.1080/08927014.2019.1642334>
- Inamdar A, Cherukattu J, Anand A, Kandasubramanian B (2018) Thermoplastic toughened high temperature Cyanate esters and its application in advanced composites <https://doi.org/10.1021/acs.iecr.7b05202>
- International Maritime Organization (2008) International Convention on the Control of Harmful Anti-fouling Systems on Ships
- Ishimaru N, Tsukegi T, Wakisaka M et al (2012) Effects of poly(L-lactic acid) hydrolysis on attachment of barnacle cypris larvae. *Polym Degrad Stab* 97:2170–2176. <https://doi.org/10.1016/j.polymdegradstab.2012.08.012>
- ISO 14040:2006 (2006) Environmental management—life cycle assessment—principles and framework Switzerland Geneva
- Issac MN, Kandasubramanian B (2021) Effect of microplastics in water and aquatic systems. *Environ Sci Pollut Res* 28:19544–19562. <https://doi.org/10.1007/s11356-021-13184-2>
- Ista LK, Callow ME, Finlay JA et al (2004) Effect of substratum surface chemistry and surface energy on attachment of marine bacteria and algal spores. *Appl Environ Microbiol* 70:4151–4157. <https://doi.org/10.1128/AEM.70.7.4151-4157.2004>
- Jain V, Kandasubramanian B (2020) Functionalized graphene materials for hydrogen storage. *J Mater Sci* 55:1865–1903. <https://doi.org/10.1007/s10853-019-04150-y>
- Jaya Prakash N, Kandasubramanian B (2021) Nanocomposites of MXene for industrial applications. *J Alloys Compd* 862 <https://doi.org/10.1016/j.jallcom.2020.158547>
- Jayalakshmi CG, Inamdar A, Anand A, Kandasubramanian B (2019) Polymer matrix composites as broadband radar absorbing structures for stealth aircrafts. *J. Appl. Polym. Sci.* 136
- Johnson AL, Bergman J, Sjögren M, Bohlin L (2004) Synthesis of baretin. *Tetrahedron* 60:961–965. <https://doi.org/10.1016/j.tet.2003.11.031>
- Jotun Group (2021) Jotun group report 2019. http://emag.allegro.no/jotun/2019/groupreport/page_57.html. Accessed 24 Jan 2022
- Kamatou GP, Vermaak I, Viljoen AM (2012) Eugenol—from the remote Maluku Islands to the international market place: a review of a remarkable and versatile molecule. *Molecules* 17:6953–6981. <https://doi.org/10.3390/molecules17066953>
- Kébir N, Campistron I, Laguerre A et al (2007) Use of telechelic cis-1,4-polyisoprene cationomers in the synthesis of antibacterial ionic polyurethanes and copolyurethanes bearing ammonium groups. *Biomaterials* 28:4200–4208. <https://doi.org/10.1016/j.biomaterials.2007.06.006>
- Kenawy ER, Worley SD, Broughton R (2007) The chemistry and applications of antimicrobial polymers: a state-of-the-art review. *Biomacromol* 8:1359–1384. <https://doi.org/10.1021/bm061150q>
- Kiil S, Weinell CE, Yebra DM, Dam-Johansen K (2006) Marine biofouling protection: Design of controlled release antifouling paints. *AIChE Annu Meet Conf Proc* 181–238
- Kobori H, Sullivan CW, Shizuya H (1984) Bacterial plasmids in antarctic natural microbial assemblages. *Appl Environ Microbiol* 48:515–518. <https://doi.org/10.1128/aem.48.3.515-518.1984>
- Korde JM, Sreekumar A V., Kandasubramanian B (2020) Corrosion inhibition of 316L-type stainless steel under marine environments using epoxy/waste plastic soot coatings. *SN Appl Sci* 2 <https://doi.org/10.1007/s42452-020-3096-2>
- Kotrikla A (2009) Environmental management aspects for TBT antifouling wastes from the shipyards. *J Environ Manage* 90:77–85. <https://doi.org/10.1016/j.jenvman.2008.07.017>
- Kumar CV, Kandasubramanian B (2019) Advances in ablative composites of carbon based materials: a review. *Ind Eng Chem Res* 58:22663–22701. <https://doi.org/10.1021/acs.iecr.9b04625>
- Kwong TFN, Miao L, Li X, Qian PY (2006) Novel antifouling and antimicrobial compound from a marine-derived fungus *Ampelomyces* sp. *Mar Biotechnol* 8:634–640. <https://doi.org/10.1007/s10126-005-6146-2>
- Langlois V, Vallee-Rehel K, Peron JJ et al (2002) Synthesis and hydrolytic degradation of graft copolymers containing poly(lactic acid) side chains: In vitro release studies of bioactive molecules. *Polym Degrad Stab* 76:411–417. [https://doi.org/10.1016/S0141-3910\(02\)00042-3](https://doi.org/10.1016/S0141-3910(02)00042-3)
- Lejars M, Margaillan A, Bressy C (2012) Fouling release coatings: a nontoxic alternative to biocidal antifouling coatings. *Chem Rev* 112:4347–4390. <https://doi.org/10.1021/cr200350v>
- Li Y, Gao YH, Li XS et al (2010) Influence of surface free energy on the adhesion of marine benthic diatom *Nitzschia closterium* MMDL533. *Colloids Surfaces B Biointerfaces* 75:550–556. <https://doi.org/10.1016/j.colsurfb.2009.09.026>
- Limna Mol VP, Raveendran TV, Parameswaran PS (2009) Antifouling activity exhibited by secondary metabolites of the marine sponge, *Haliclona exigua* (Kirkpatrick). *Int Biodeterior Biodegrad* 63:67–72. <https://doi.org/10.1016/j.ibiod.2008.07.001>
- Liu Y, Zhao Q (2005) Influence of surface energy of modified surfaces on bacterial adhesion. *Biophys Chem* 117:39–45. <https://doi.org/10.1016/j.bpc.2005.04.015>
- López-Malo A, Alzamora SM, Palou E (2002) *Aspergillus flavus* dose-response curves to selected natural and synthetic antimicrobials. *Int J Food Microbiol* 73:213–218. [https://doi.org/10.1016/S0168-1605\(01\)00639-0](https://doi.org/10.1016/S0168-1605(01)00639-0)
- Macken A, Giltrap M, Ryall K et al (2009) A test battery approach to the ecotoxicological evaluation of cadmium and copper employing a battery of marine bioassays. *Ecotoxicology* 18:470–480. <https://doi.org/10.1007/s10646-009-0305-6>

- Majumdar P, Lee E, Patel N et al (2008) Combinatorial materials research applied to the development of new surface coatings IX: an investigation of novel antifouling/fouling-release coatings containing quaternary ammonium salt groups. *Biofouling* 24:185–200. <https://doi.org/10.1080/08927010801894660>
- Maréchal JP, Hellio C (2009) Challenges for the development of new non-toxic antifouling solutions. *Int J Mol Sci* 10:4623–4637. <https://doi.org/10.3390/ijms10114623>
- Maréchal JP, Hellio C (2011) Antifouling activity against barnacle cypris larvae: do target species matter (*Amphibalanus amphitrite* versus *Semibalanus balanoides*)? *Int Biodeterior Biodegrad* 65:92–101. <https://doi.org/10.1016/j.ibiod.2010.10.002>
- Marris E (2006) Marine natural products: drugs from the deep. *Nature* 443:904–905. <https://doi.org/10.1038/443904a>
- Mizerska U, Fortuniak W, Chojnowski J et al (2009) Polysiloxane cationic biocides with imidazolium salt (ImS) groups, synthesis and antibacterial properties. *Eur Polym J* 45:779–787. <https://doi.org/10.1016/j.eurpolymj.2008.11.045>
- Mohandes F, Salavati-Niasari M (2014) In vitro comparative study of pure hydroxyapatite nanorods and novel polyethylene glycol/graphene oxide/hydroxyapatite nanocomposite. *J Nanoparticle Res* 16. <https://doi.org/10.1007/s11051-014-2604-y>
- Mokrini R, Ben MM, Daoudi M et al (2008) Meroditerpenoids and derivatives from the brown alga *Cystoseira baccata* and their antifouling properties. *J Nat Prod* 71:1806–1811. <https://doi.org/10.1021/np8004216>
- Mudryk ZJ (2002) Antibiotic resistance among bacteria inhabiting surface and subsurface water layers in estuarine Lake Gardno. *Polish J Environ Stud* 11:401–406
- Cole NP, Kuznetsov ISA et al (2021) Experimental and DFT studies of gadolinium decorated graphene oxide materials for their redox properties and as a corrosion inhibition barrier layer on Mg AZ13 alloy in a 3.5% NaCl environment. *RSC Adv* 11:22095–22105. <https://doi.org/10.1039/D1RA03495B>
- Nakano D, Strayer DL (2014) Biofouling animals in fresh water: biology, impacts, and ecosystem engineering. *Front Ecol Environ* 12:167–175. <https://doi.org/10.1890/130071>
- Nandakumar K, Obika H, Shinozaki T et al (2002) Impact of pulsed laser irradiations from Nd:YAG laser on the larvae of the fouling barnacle *Balanus amphitrite*. *Biofouling* 18:257–262. <https://doi.org/10.1080/08927010290032403>
- National Oceanic and Atmospheric Administration (2020) What are barnacles? In: Natl. Ocean Serv. - U.S. Dep. Commer. <https://oceanservice.noaa.gov/facts/barnacles.html>. Accessed 15 Oct 2020
- Nurdin N, Helary G, Sauvet G (1993a) Biocidal polymers active by contact. III. Ageing of biocidal polyurethane coatings in water. *J Appl Polym Sci* 50:671–678. <https://doi.org/10.1002/app.1993.070500412>
- Nurdin N, Helary G, Sauvet G (1993b) Biocidal polymers active by contact. II. Biological evaluation of polyurethane coatings with pendant quaternary ammonium salts. *J Appl Polym Sci* 50:663–670. <https://doi.org/10.1002/app.1993.070500411>
- Okamura H, Nishida T, Ono Y, Shim WJ (2003) Phytotoxic effects of antifouling compounds on nontarget plant species. *Bull Environ Contam Toxicol* 71:881–886
- Okoro HK, Fatoki OS, Adekola FA et al (2011) Sources, environmental levels and toxicity of organotin in marine environment - A review. *Asian J Chem* 23:473–482
- Oliva M, Martinelli E, Galli G, Pretti C (2017) PDMS-based films containing surface-active amphiphilic block copolymers to combat fouling from barnacles *B. amphitrite* and *B. improvisus*. *Polymer (guildf)* 108:476–482. <https://doi.org/10.1016/j.polymer.2016.12.021>
- Olivier-Bourbigou H, Magna L, Morvan D (2010) Ionic liquids and catalysis: recent progress from knowledge to applications. *Appl Catal A Gen* 373:1–56. <https://doi.org/10.1016/j.apcata.2009.10.008>
- Ortlepp S, Pedpradap S, Dobretsov S, Proksch P (2008) Antifouling activity of sponge-derived polybrominated diphenyl ethers and synthetic analogues. *Biofouling* 24:201–208. <https://doi.org/10.1080/08927010802008096>
- Ortlepp S, Sjögren M, Dahlström M et al (2007) Antifouling activity of bromotyrosine-derived sponge metabolites and synthetic analogues. *Mar Biotechnol* 9:776–785. <https://doi.org/10.1007/s10126-007-9029-x>
- Owen R, Knap A, Toasperm M, Carbery K (2002) Inhibition of coral photosynthesis by the antifouling herbicide Irgarol 1051. *Mar Pollut Bull* 44:623–632. [https://doi.org/10.1016/S0025-326X\(01\)00303-4](https://doi.org/10.1016/S0025-326X(01)00303-4)
- Palaniappan N, Alphonsa J, Cole IS, et al (2019) Rapid investigation expiry drug green corrosion inhibitor on mild steel in NaCl medium. *Mater Sci Eng B Solid-State Mater Adv Technol* 249. <https://doi.org/10.1016/j.mseb.2019.114423>
- Palaniappan N, Cole IS, Damodaran K et al (2020) Experimental and DFT studies of porous carbon covalently functionalized by polyaniline as a corrosion inhibition barrier on nickel-based alloys in acidic media. *RSC Adv* 10:12151–12165. <https://doi.org/10.1039/d0ra00593b>
- Parimal K, Khale A, Pramod K (2011) Resins from herbal origin and a focus on their applications. *Int J Pharm Sci Res* 2:1077–1085
- Pawar SS, Baloji Naik R, Rath SK et al (2019) Photoinduced hydrophilicity and self-cleaning characteristics of silicone-modified soya alkyl/TiO₂ nanocomposite coating. *J Coatings Technol Res*. <https://doi.org/10.1007/s11998-019-00253-y>
- Pawar SS, Baloji Naik R, Rath SK et al (2020) Photoinduced hydrophilicity and self-cleaning characteristics of silicone-modified soya alkyl/TiO₂ nanocomposite coating. *J Coatings Technol Res* 17:719–730. <https://doi.org/10.1007/s11998-019-00253-y>
- Pérez M, García M, Blustein G (2015) Evaluation of low copper content antifouling paints containing natural phenolic compounds as bioactive additives. *Mar Environ Res* 109:177–184. <https://doi.org/10.1016/j.marenvres.2015.07.006>
- Pérez M, García M, Blustein G, Stupak M (2007) Tannin and tannate from the quebracho tree: An eco-friendly alternative for controlling marine biofouling. *Biofouling* 23:151–159. <https://doi.org/10.1080/08927010701189484>
- Phang IY, Aldred N, Clare AS, Vancso GJ (2008) Towards a nanomechanical basis for temporary adhesion in barnacle cyprids (*Semibalanus balanoides*). *J R Soc Interface* 5:397–401. <https://doi.org/10.1098/rsif.2007.1209>
- PPG Industries UK Ltd (2019) PPG 2019 Sustainability Report - Emissions
- Price RR, Patchan M, Clare A et al (1992) Performance enhancement of natural antifouling compounds and their analogs through microencapsulation and controlled release. *Biofouling* 6:207–216. <https://doi.org/10.1080/08927019209386223>
- Qi SH, Miao L, Gao CH et al (2010) New steroids and a new alkaloid from the gorgonian *Isis minorbrachyblasta*: Structures, cytotoxicity, and antilarval activity. *Helv Chim Acta* 93:511–516. <https://doi.org/10.1002/hlca.200900250>
- Qi SH, Zhang S, Qian PY et al (2006) Ten new antifouling briarane diterpenoids from the South China Sea gorgonian *Juncella juncea*. *Tetrahedron* 62:9123–9130. <https://doi.org/10.1016/j.tet.2006.07.049>
- Qi SH, Zhang S, Qian PY, Wang BG (2008a) Antifeedant, antibacterial, and antilarval compounds from the South China Sea seagrass *Enhalus acoroides*. *Bot Mar* 51:441–447. <https://doi.org/10.1515/BOT.2008.054>
- Qi SH, Zhang S, Qian PY, Xu HH (2009) Antifeedant and antifouling briaranes from the South China Sea gorgonian *Juncella juncea*. *Chem Nat Compd* 45:49–54. <https://doi.org/10.1007/s10600-009-9255-8>

- Qi SH, Zhang S, Yang LH, Qian PY (2008b) Antifouling and antibacterial compounds from the gorgonians *Subergorgia suberosa* and *Scrippearia gracilis*. *Nat Prod Res* 22:154–166. <https://doi.org/10.1080/14786410701642441>
- Qian PY, Xu Y, Fusetani N (2009) Natural products as antifouling compounds: recent progress and future perspectives. *Biofouling* 26:223–234. <https://doi.org/10.1080/08927010903470815>
- Qiu Y, Deng ZW, Xu M et al (2008) New A-nor steroids and their antifouling activity from the Chinese marine sponge *Acanthella cavernosa*. *Steroids* 73:1500–1504. <https://doi.org/10.1016/j.steroids.2008.08.006>
- Rajitha K, Nanchaiah YV, Venugopalan VP (2020) Insight into bacterial biofilm-barnacle larvae interactions for environmentally benign antifouling strategies. *Int Biodeterior Biodegrad* 149:104937. <https://doi.org/10.1016/j.ibiod.2020.104937>
- Ralston E, Swain G (2009) Bioinspiration—the solution for biofouling control? *Bioinspir Biomim* 4:015007. <https://doi.org/10.1088/1748-3182/4/1/015007>
- Rath SK, Chavan JG, Ghorpade TK et al (2018) Structure–property correlations of foul release coatings based on low hard segment content poly(dimethylsiloxane–urethane–urea). *J Coatings Technol Res* 15:185–198. <https://doi.org/10.1007/s11998-017-9982-x>
- Rath SK, Chavan JG, Sasane S et al (2009) Coatings of PDMS-modified epoxy via urethane linkage: Segmental correlation length, phase morphology, thermomechanical and surface behavior. *Prog Org Coatings* 65:366–374. <https://doi.org/10.1016/j.porgcoat.2009.02.007>
- Rath SK, Chavan JG, Sasane S et al (2010) Two component silicone modified epoxy foul release coatings: effect of modulus, surface energy and surface restructuring on pseudobarnacle and macrofouling behavior. *Appl Surf Sci* 256:2440–2446. <https://doi.org/10.1016/j.apsusc.2009.10.084>
- Ravikumar T, Murata H, Koepsel RR, Russell AJ (2006) Surface-active antifungal polyquaternary amine. *Biomacromol* 7:2762–2769. <https://doi.org/10.1021/bm060476w>
- Reddy GKK, Nanchaiah YV, Venugopalan VP (2017) Long alkyl-chain imidazolium ionic liquids: antibiofilm activity against phototrophic biofilms. *Colloids Surfaces B Biointerfaces* 155:487–496. <https://doi.org/10.1016/j.colsurfb.2017.04.040>
- Rittschof D, Chai C, Lay S, et al (2015) وإدارته املس تقبل ية: وجه نظر 4. 20:16–23
- Rittschof D, Costlow J (1989) Bryozoan and barnacle settlement in relation to initial surface wettability: a comparison of laboratory and field studies. *Sci Mar* 53:411–416
- Rodgman A, Perfetti TA (2008) The chemical components of tobacco and tobacco smoke. CRC Press
- Rogers RD, Seddon KR (2003) Ionic liquids—solvents of the future? *Science* (80-) 302:792–793. <https://doi.org/10.1126/science.1090313>
- Rouhi AM (1998) Former EPA adviser voices doubts over regulations restricting antifouling paints. 41–42
- Ruberto G, Baratta MT (2000) Antioxidant activity of selected essential oil components in two lipid model systems. *Food Chem* 69:167–174. [https://doi.org/10.1016/S0308-8146\(99\)00247-2](https://doi.org/10.1016/S0308-8146(99)00247-2)
- Rudawska A, Jacniacka E (2009) Analysis for determining surface free energy uncertainty by the Owen-Wendt method. *Int J Adhes Adhes* 29:451–457. <https://doi.org/10.1016/j.jadhadh.2008.09.008>
- Santos MM, Ten Hallers-Tjabbes CC, Santos AM, Vieira N (2002) Imposen in *Nucella lapillus*, a bioindicator for TBT contamination: Re-survey along the Portuguese coast to monitor the effectiveness of EU regulation. *J Sea Res* 48:217–223. [https://doi.org/10.1016/S1385-1101\(02\)00166-1](https://doi.org/10.1016/S1385-1101(02)00166-1)
- Sapozhnikova Y, Wirth E, Schiff K, Fulton M (2013) Antifouling biocides in water and sediments from California marinas. *Mar Pollut Bull* 69:189–194. <https://doi.org/10.1016/j.marpolbul.2013.01.039>
- Sarkar P, Kandasubramanian B (2021a) Fluorine-free superhydrophobic characterized coatings: A mini review. *Marit Technol Res* 3:365–376. <https://doi.org/10.33175/mtr.2021.251814>
- Sarkar PK, Kandasubramanian B (2021b) Metals to polymer composites for submerged hull: a paradigm shift. *Polym Technol Mater*. <https://doi.org/10.1080/25740881.2021.1930048>
- Schultz MP (2004) Frictional resistance of antifouling coating systems. *J Fluids Eng Trans ASME* 126:1039–1047. <https://doi.org/10.1115/1.1845552>
- Schultz MP (2007) Effects of coating roughness and biofouling on ship resistance and powering. *Biofouling* 23:331–341. <https://doi.org/10.1080/08927010701461974>
- Schultz MP, Bendick JA, Holm ER, Hertel WM (2011) Economic impact of biofouling on a naval surface ship. *Biofouling* 27:87–98. <https://doi.org/10.1080/08927014.2010.542809>
- Schultz MP, Swain GW (2000) The influence of biofilms on skin friction drag. *Biofouling* 15:129–139. <https://doi.org/10.1080/08927010009386304>
- Selim MS, Shenashen MA, El-Safty SA et al (2017) Recent progress in marine foul-release polymeric nanocomposite coatings. *Prog Mater Sci* 87:1–32. <https://doi.org/10.1016/j.pmatsci.2017.02.001>
- Shivapooja P, Cao C, Orihuela B et al (2016) Incorporation of silicone oil into elastomers enhances barnacle detachment by active surface strain. *Biofouling* 32:1017–1028. <https://doi.org/10.1080/08927014.2016.1209186>
- Singer IL, Kohl JG, Patterson M (2000) Mechanical aspects of silicone coatings for hard foulant control. *Biofouling* 16:301–309. <https://doi.org/10.1080/08927010009378453>
- Sjögren M, Dahlström M, Göransson U et al (2004a) Recruitment in the field of *Balanus improvisus* and *Mytilus edulis* in response to the antifouling cyclopeptides baretin and 8,9-dihydrobaretin from the marine sponge *Geodia barretti*. *Biofouling* 20:291–297. <https://doi.org/10.1080/08927010400027027>
- Sjögren M, Göransson U, Johnson AL et al (2004b) Antifouling activity of brominated cyclopeptides from the marine sponge *Geodia barretti*. *J Nat Prod* 67:368–372. <https://doi.org/10.1021/np0302403>
- Sjögren M, Johnson AL, Hedner E et al (2006) Antifouling activity of synthesized peptide analogs of the sponge metabolite baretin. *Peptides* 27:2058–2064. <https://doi.org/10.1016/j.peptides.2006.03.027>
- Sjollema SB, Martínezgarcía G, Van Der Geest HG et al (2014) Hazard and risk of herbicides for marine microalgae. *Environ Pollut* 187:106–111. <https://doi.org/10.1016/j.envpol.2013.12.019>
- Skattebøl L, Nilsen NO, Stenström Y et al (2006) The antifouling activity of some juvenoids on three species of acorn barnacle, *Balanus*. *Pest Manag Sci* 62:610–616. <https://doi.org/10.1002/ps.1210>
- Song S, Demirel YK, Atlar M (2019) An investigation into the effect of biofouling on the ship hydrodynamic characteristics using CFD. *Ocean Eng* 175:122–137. <https://doi.org/10.1016/j.oceaneng.2019.01.056>
- Srinivasan M, Swain GW (2007) Managing the use of copper-based antifouling paints. *Environ Manage* 39:423–441. <https://doi.org/10.1007/s00267-005-0030-8>
- Stein J, Truby K, Wood CD et al (2003) Silicone foul release coatings: effect of the interaction of oil and coating functionalities on the magnitude of macrofouling attachment strengths. *Biofouling* 19:71–82. <https://doi.org/10.1080/0892701031000089525>
- Strand J, Jacobsen JA (2005) Accumulation and trophic transfer of organotins in a marine food web from the Danish coastal waters. *Sci Total Environ* 350:72–85. <https://doi.org/10.1016/j.scitotenv.2005.02.039>
- Stupak ME, García MT, Pérez MC (2003) Non-toxic alternative compounds for marine antifouling paints. *Int Biodeterior Biodegrad* 52:49–52. [https://doi.org/10.1016/S0964-8305\(03\)00035-0](https://doi.org/10.1016/S0964-8305(03)00035-0)

- Suraj Belgaonkar M, Kandasubramanian B (2021) Hyperbranched polymer-based nanocomposites: synthesis, progress, and applications. *Eur Polym J* 147:<https://doi.org/10.1016/j.eurpolymj.2021.110301>
- Swain GW, Kovach B, Touzot A et al (2007) Measuring the performance of today's antifouling coatings. *J Sh Prod* 23:164–170
- Tasso M, Pettitt ME, Cordeiro AL et al (2009) Antifouling potential of subtilisin a immobilized onto maleic anhydride copolymer thin films. *Biofouling* 25:505–516. <https://doi.org/10.1080/08927010902930363>
- Telegdi J, Trif L, Románszki L (2016) Smart anti-biofouling composite coatings for naval applications. *Smart Compos Coatings Membr Transp Struct Environ Energy Appl* 123–155. <https://doi.org/10.1016/B978-1-78242-283-9.00005-1>
- Tepe B, Sokmen M, Akpulat HA et al (2005) Antioxidative activity of the essential oils of *Thymus sipyleus* subsp. *sipyleus* var. *sipyleus* and *Thymus sipyleus* subsp. *sipyleus* var. *rosulans*. *J Food Eng* 66:447–454. <https://doi.org/10.1016/j.jfoodeng.2004.04.015>
- Thakur A, Gharde S, Kandasubramanian B (2019) Electroless nickel fabrication on surface modified magnesium substrates. *Def Technol*. <https://doi.org/10.1016/j.dt.2019.04.006>
- Thiyagarajan VVP, Nair KVK, Subramoniam T (1999) Fouling barnacles: larval development, settlement behavior and control technology. *J Indian Inst Sci* 79:399–414
- Thomas KV, Brooks S (2010) The environmental fate and effects of antifouling paint biocides. *Biofouling* 26:73–88. <https://doi.org/10.1080/08927010903216564>
- Todd JS, Zimmerman RC, Crews P, Alberte RS (1993) The antifouling activity of natural and synthetic phenolic acid sulphate esters. *Phytochemistry* 34:401–404. [https://doi.org/10.1016/0031-9422\(93\)80017-M](https://doi.org/10.1016/0031-9422(93)80017-M)
- Tokiwa Y, Calabria BP (2007) Biodegradability and biodegradation of polyesters. *J Polym Environ* 15:259–267. <https://doi.org/10.1007/s10924-007-0066-3>
- Truby K, Wood C, Stein J et al (2000) Evaluation of the performance enhancement of silicone biofouling-release coatings by oil incorporation. *Biofouling* 15:141–150. <https://doi.org/10.1080/08927010009386305>
- Tsoukatou M, Maréchal JP, Hellio C et al (2007) Evaluation of the activity of the sponge metabolites avarol and avarone and their synthetic derivatives against fouling micro- and macroorganisms. *Molecules* 12:1022–1034. <https://doi.org/10.3390/12051022>
- Tsukamoto S, Kato H, Hirota H, Fusetani N (1997) Antifouling terpenes and steroids against barnacle larvae from marine sponges. *Biofouling* 11:283–291. <https://doi.org/10.1080/08927019709378337>
- Turan O, Demirel YK, Day S, Tezdogan T (2016) Experimental determination of added hydrodynamic resistance caused by marine biofouling on ships. *Transp Res Procedia* 14:1649–1658. <https://doi.org/10.1016/j.trpro.2016.05.130>
- Turner A, Singh N, Richards JP (2009) Bioaccessibility of metals in soils and dusts contaminated by marine antifouling paint particles. *Environ Pollut* 157:1526–1532. <https://doi.org/10.1016/j.envpol.2009.01.008>
- Uzun D, Ozyurt R, Demirel YK, Turan O (2020) Does the barnacle settlement pattern affect ship resistance and powering? *Appl Ocean Res* 95:<https://doi.org/10.1016/j.apor.2019.102020>
- Vekariya RL (2017) A review of ionic liquids: applications towards catalytic organic transformations. *J Mol Liq* 227:44. <https://doi.org/10.1016/j.molliq.2016.11.123>
- Venkata Nanchaiah Y, Reddy GKK, Lalithamanasa P, Venugopalan VP (2012) The ionic liquid 1-alkyl-3-methylimidazolium demonstrates comparable antimicrobial and antibiofilm behavior to a cationic surfactant. *Biofouling* 28:1141–1149. <https://doi.org/10.1080/08927014.2012.736966>
- Villa F, Albanese D, Giussani B et al (2010) Hindering biofilm formation with zosteric acid. *Biofouling* 26:739–752. <https://doi.org/10.1080/08927014.2010.511197>
- Villa F, Pitts B, Stewart PS et al (2011) Efficacy of zosteric acid sodium salt on the yeast biofilm model *Candida albicans*. *Microb Ecol* 62:584–598. <https://doi.org/10.1007/s00248-011-9876-x>
- Villa F, Remelli W, Forlani F et al (2012) Altered expression level of *Escherichia coli* proteins in response to treatment with the antifouling agent zosteric acid sodium salt. *Environ Microbiol* 14:1753–1761. <https://doi.org/10.1111/j.1462-2920.2011.02678.x>
- Vo Dong PA, Azzaro-Pantel C, Cadene AL (2018) Economic and environmental assessment of recovery and disposal pathways for CFRP waste management. *Resour Conserv Recycl* 133:63–75. <https://doi.org/10.1016/j.resconrec.2018.01.024>
- Witthuhn RC, van der Merwe E, Venter P, Cameron M (2012) Guaiacol production from ferulic acid, vanillin and vanillic acid by *Alicyclobacillus acidoterrestris*. *Int J Food Microbiol* 157:113–117. <https://doi.org/10.1016/j.ijfoodmicro.2012.04.022>
- World Resources Institute (2020) Greenhouse gas protocol—Product life cycle accounting reporting standard
- Xiong H, Song L, Xu Y et al (2007) Characterization of proteolytic bacteria from the Aleutian deep-sea and their proteases. *J Ind Microbiol Biotechnol* 34:63–71. <https://doi.org/10.1007/s10295-006-0165-5>
- Xu Q, Barrios CA, Cutright T, Newby BMZ (2005) Evaluation of toxicity of capsaicin and zosteric acid and their potential application as antifoulants. *Environ Toxicol* 20:467–474. <https://doi.org/10.1002/tox.20134>
- Xu Y, He H, Schulz S et al (2010) Potent antifouling compounds produced by marine Streptomyces. *Bioresour Technol* 101:1331–1336. <https://doi.org/10.1016/j.biortech.2009.09.046>
- Yebra DM, Kiil S, Dam-Johansen K (2004) Antifouling technology—past, present and future steps towards efficient and environmentally friendly antifouling coatings. *Prog Org Coatings* 50:75–104. <https://doi.org/10.1016/j.porgcoat.2003.06.001>
- Zabka M, Pavela R (2013) Antifungal efficacy of some natural phenolic compounds against significant pathogenic and toxinogenic filamentous fungi. *Chemosphere* 93:1051–1056. <https://doi.org/10.1016/j.chemosphere.2013.05.076>
- Zhou L, Zheng H, Tang Y et al (2013) Eugenol inhibits quorum sensing at sub-inhibitory concentrations. *Biotechnol Lett* 35:631–637. <https://doi.org/10.1007/s10529-012-1126-x>
- Zhou X, Zhang Z, Xu Y et al (2009) Flavone and isoflavone derivatives of terrestrial plants as larval settlement inhibitors of the barnacle *Balanus amphitrite*. *Biofouling* 25:69–76. <https://doi.org/10.1080/08927010802455941>

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